

ABSTRACT

Casting a vote is not only the right of a citizen but also a responsibility of the citizen. The citizens of a country get an opportunity to vote for their representatives who will represent the needs and suggestions of citizens. These representatives are elected through the process called elections. The traditional electoral system necessitates the actual presence of the voter which causes discomfort to the physically challenged people. Also, there are chances of vote tampering. This paper discusses the proposed solution that will solve the above problems. Our proposed solution is to use an **Online Voting System** using **Ethereum Blockchain**. This web-based voting system helps the voters to vote from any location. The system validates the voters with the help of their Aadhar cards linked with their voter id. After which the system requires the voters to scan their face and fingerprint which will be verified with the data in the database. Blockchain technology encrypts the vote and thus it prevents every vote from tampering. It makes sure that a voter can vote only once for one candidate. The system fetches the election results quickly and thus reduces the labour cost and counting errors.

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CHAPTER 1

INTRODUCTION

As voting is an important aspect of a country's decisionmaking process, most elections and voting systems begin with a provisional ballot method. In this paper, the system validates the voter with the help of OTP which is sent to the mobile number linked with the Aadhar for verification and biometrics verification authorized by the Unique Identification Authority of India. The voting process and the data it includes are protected and saved using blockchain technology.

This system must ensure the secrecy and the security of the data and votes. Blockchain technology plays a vital role in maintaining security and encryption. When a voter casts their vote, the Blockchain infrastructure makes extremely difficult for a deceptive node to publish or repost data because it will be checked and approved by the majority of other nodes linked to this network. Smart contracts are executed after the voter casts his vote and further the voter will be unable to participate in the process. If any of the nodes want to make any changes it should attain 51% majority votes from the remaining nodes which are near to impossible.

Only when a large number of people partake in the voting process, a successful voting system will be achieved. To make them a reality, the voting technology should be userfriendly and not require the voters to queue outside the polling booth. Many approaches and concepts have been implemented and copied throughout the years to create a functioning voting system. The main purpose of this paper is to encourage more people to vote remotely, regardless of where they are or which smartphone they are using, to save time consumption and make voting more flexible and practicable for people.

CHAPTER 2

LITERATURE SURVEY

TITLE: Smart Online Voting System.

ABSTRACT: To create a safe, Face recognition-based internet voting system that attempts to address all of the flaws that currently exist in voting systems. The suggested system has several significant characteristics, including correctness and efficiency. Verifiability, ease of use, and so forth. There is no requirement for this system a poll worker, a paper ballot, or any type of electronic voting machine. Face scanners and an internet connection are the only requirements. Voters can vote from any secure location.

TITLE: Online Voting System using Cloud

ABSTRACT: Creative tickets, stunning agenda highlights, vote counting, classification, and disclosure are all included in the Online Voting Platform. These skills are pre-programmed and should not be given away on the spur of the moment, Sent an internal memo to faculty. It also allows for the creation of new ideas. Organizes polls to prevent voters from casting illegitimate ballots. They shouldn't be counted, and they shouldn't be double-checked. The simplest and most convenient method of voting is through an online platform. Both directors and voters will benefit from this method. Directors should be aware of Setting up a ticket and making a decision is possible if you follow these steps simple and practical.

TITLE: Electronic Voting based on Virtual ID of Aadhar using Blockchain Technology

ABSTRACT: Voting is one of the processes that allow citizens to find their place in society, as well as one of the rights to elect a worthy and modest leader. There are a lot of voting systems that aren't secure, therefore by incorporating the blockchain, security is ensured. Verification of Aadhar utilizing VID, and fingerprint data is used to create a digital signature has a significant role to play. **TITLE:** Smart and Secure Voting Machine using Biometrics

ABSTRACT: After the workflow has been implemented. This will assist the voter in determining whether or not his vote has been cast. coupled with the name of the candidate for whom they voted. In addition to that only the registered phone number will receive a message. will be contacted. **TITLE:** Aadhar Card Verification Base Online Polling

ABSTRACT: This voting technique, allows everyone to vote. The degree of ballot casting should be determined via web-based voting. It isn't required tallying by hand. The effect is evident, transparent, and rapid. The newly developed technology is capable of resolving a wide range of issues of the current system. The Aadhar token is used to authenticate the user. The protection of

the upgraded framework is increased. Casting can be done in a variety of ways, all of which are equally safe. The Aadhar card will be used to provide a unique identification card to ensure voter security. The break is kept to a minimum.

TITLE: Implementing Electronic Voting System With Blockchain Technology.

ABSTRACT: Blockchain can make E-Voting more secure and efficient. The key to making the user comfortable and removing the barrier between the voter and voting is security and convenience. The blockchain as a technology is made feasible through technology. Not only does blockchain have the potential to revolutionize the financial world, but also can make the process more digitally visible. It will be more secure if it is immutable. The hyperledger-based blockchain-based electronic voting system ensures users' security throughout elections privacy. When Ethereum and hyper ledger are compared. Hyperledger is more efficient than Ethereum in the majority of performance indicators.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing system needs the actual presence of the voter which is not convenient and chances of vote tampering is more. Also, the transparency of the entire system is not maintained. Our system uses blockchain technology which helps in maintaining the transparency, accuracy, verifiability, ease, security, of the system. The suggested method uses double verification for increased security. Because the system is an online application, users may vote from anywhere. The voter does not need to go to a polling location to vote, allowing a larger number of residents to participate. The Ethereum Blockchain will be used to safeguard the data in the proposed system.

3.2 PROPOSED SYSTEM

Our system validates the voter by OTP and biometrics verification authorized by the Unique Identification Authority of India. The voting process and the data it includes are protected and saved using blockchain technology. Since the blockchain infrastructure makes it difficult for a deceptive node to publish or repost data, when a voter casts their vote, the data will be checked and accepted by the majority of other nodes linked to this network. After the voter has cast his vote, smart contracts are performed, and the voter will no longer be able to vote.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 FUNCTIONAL REQUIREMENTS

These are the requirements that the end user specifically demands as basic facilities that the system should offer. All these functionalities need to be necessarily incorporated into the system as a part of the contract. These are represented or stated in the form of input to be given to the system, the operation performed and the output expected. They are basically the requirements stated by the user which one can see directly in the final product, unlike the non-functional requirements. Functional requirements are specifications that define the specific behavior or functions of a system, software, or product. They describe what the system should do, its features, and how it should perform under certain conditions. In the context of software development, functional requirements serve as a blueprint for the developers, guiding them in building a system that meets the needs and expectations of the users and stakeholders.

- ELECTION
- ADMIN

4.2 NON- FUNCTIONAL REQUIREMENTS

4.2.1 HARDWARE REQUIREMENTS

- | | |
|-------------|-----------------------------|
| ➤ Processor | - Pentium –IV |
| ➤ RAM | - 4 GB (min) |
| ➤ Hard Disk | - 20 GB |
| ➤ Key Board | - Standard Windows Keyboard |
| ➤ Mouse | - Two or Three Button Mouse |
| ➤ Monitor | - SVGA |

4.2.2 SOFTWARE REQUIREMENTS

- | | |
|--------------------|--------------------------|
| ➤ Coding Language | : Python. |
| ➤ Operating system | : Windows 7 Ultimate. |
| ➤ Front-End | : Python. |
| ➤ Back-End | : Django-ORM |
| ➤ Designing | : Html, CSS, JavaScript. |
| ➤ Data Base | : MySQL. |

CHAPTER 5

SYSTEM STUDY

5.1 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

5.2 FEASIBILITY ANALYSIS

Three key considerations involved in the feasibility analysis are

- **ECONOMICAL FEASIBILITY**
- **TECHNICAL FEASIBILITY**
- **SOCIAL FEASIBILITY**

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

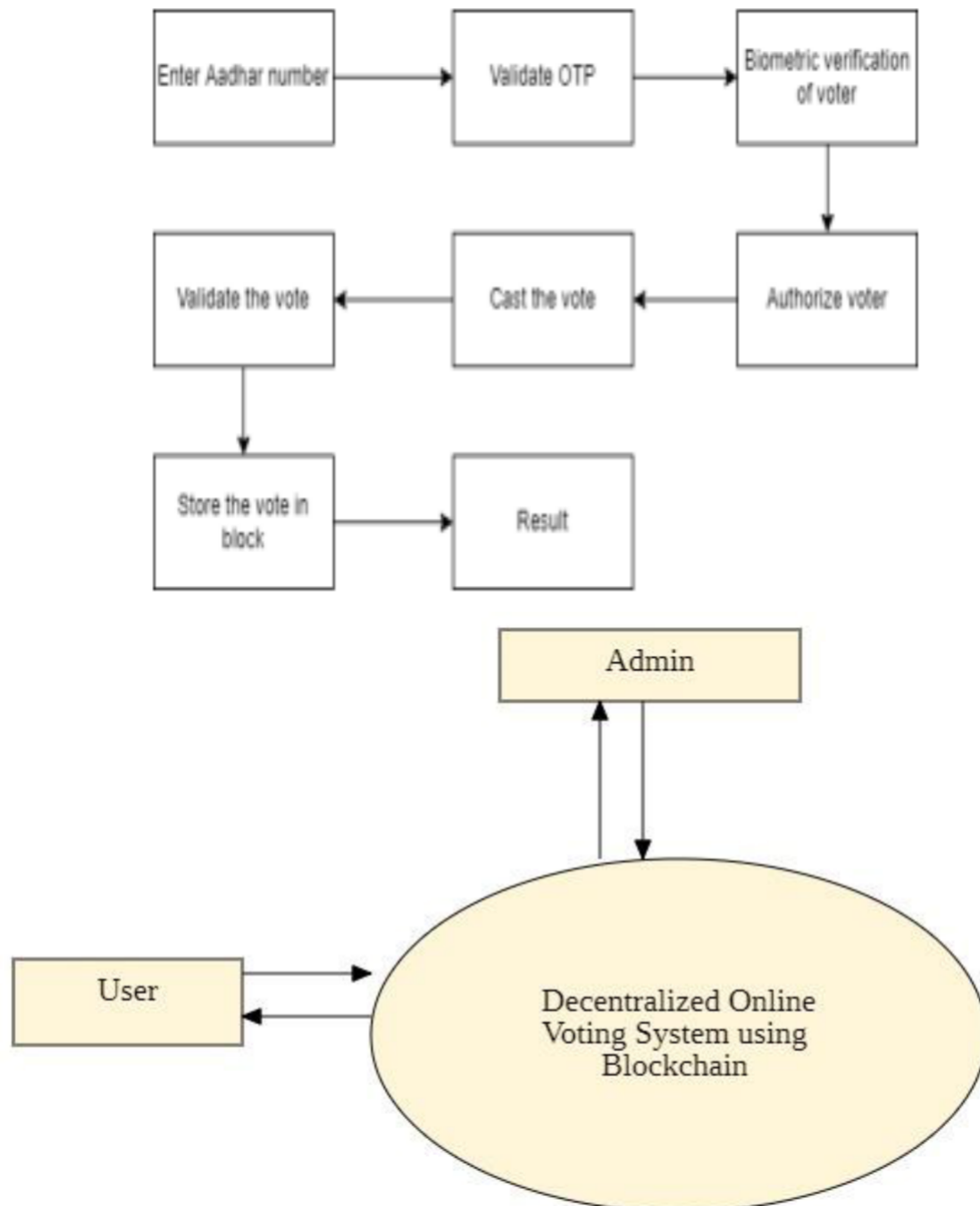
SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER 6

SYSTEM DESIGN

6.1 SYSTEM ARCHITECTURE



6.2 UML DIAGRAM'S:

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

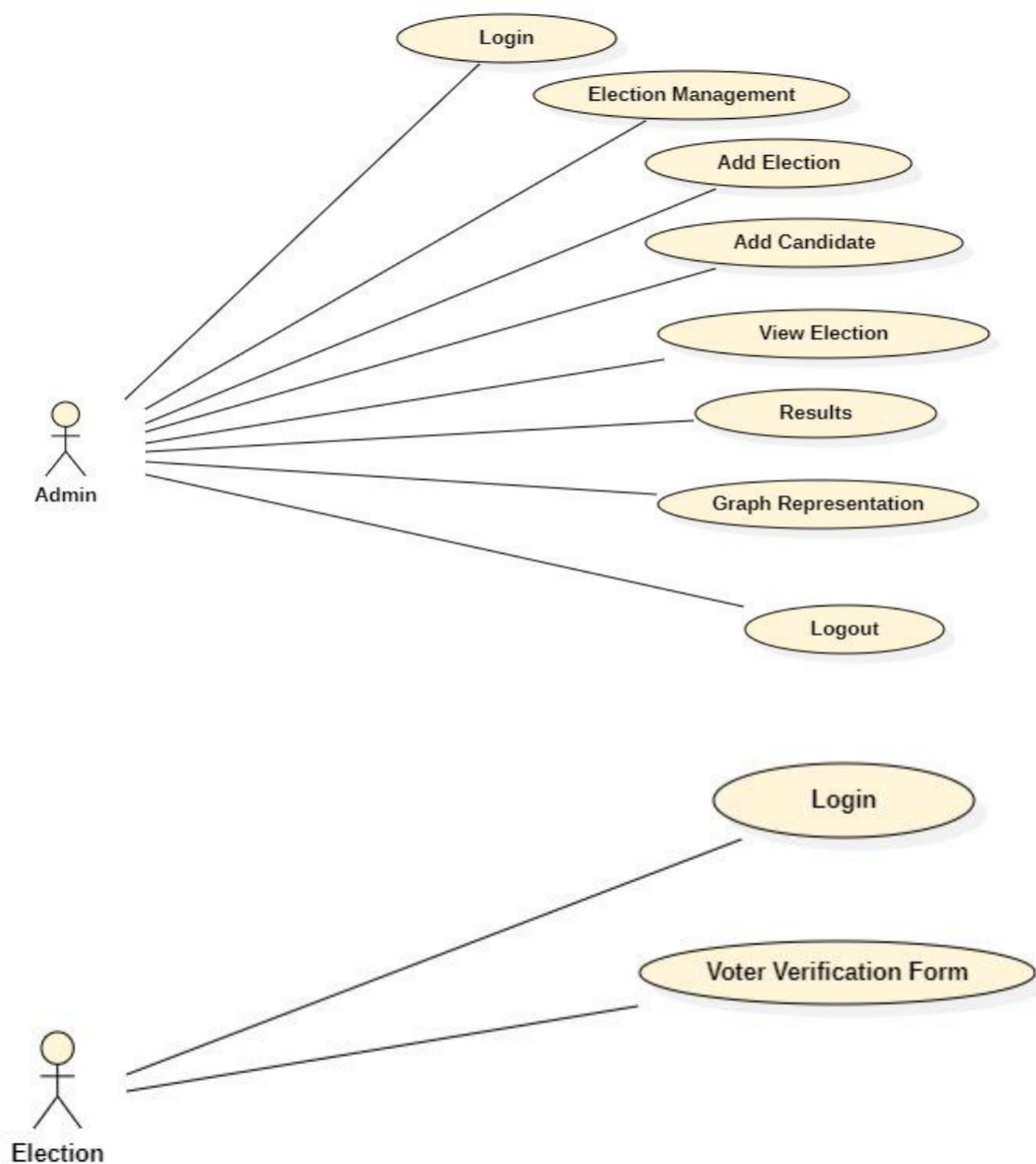
GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

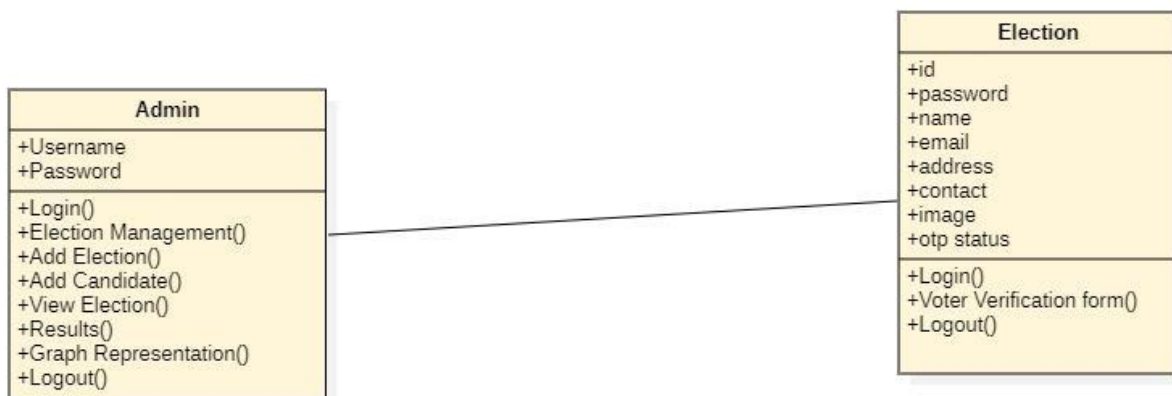
6.2.1 USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



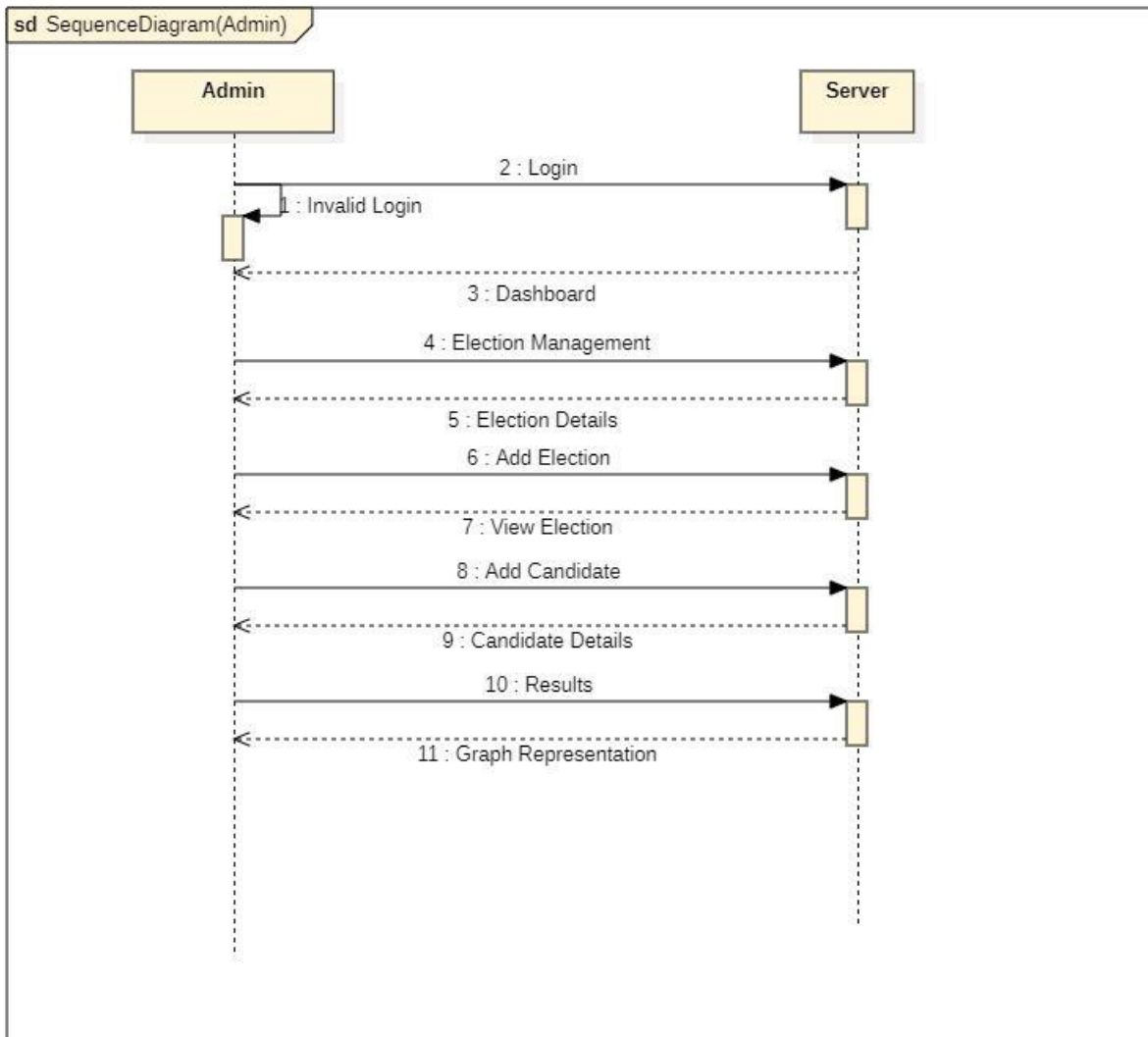
6.2.2 CLASS DIAGRAM:

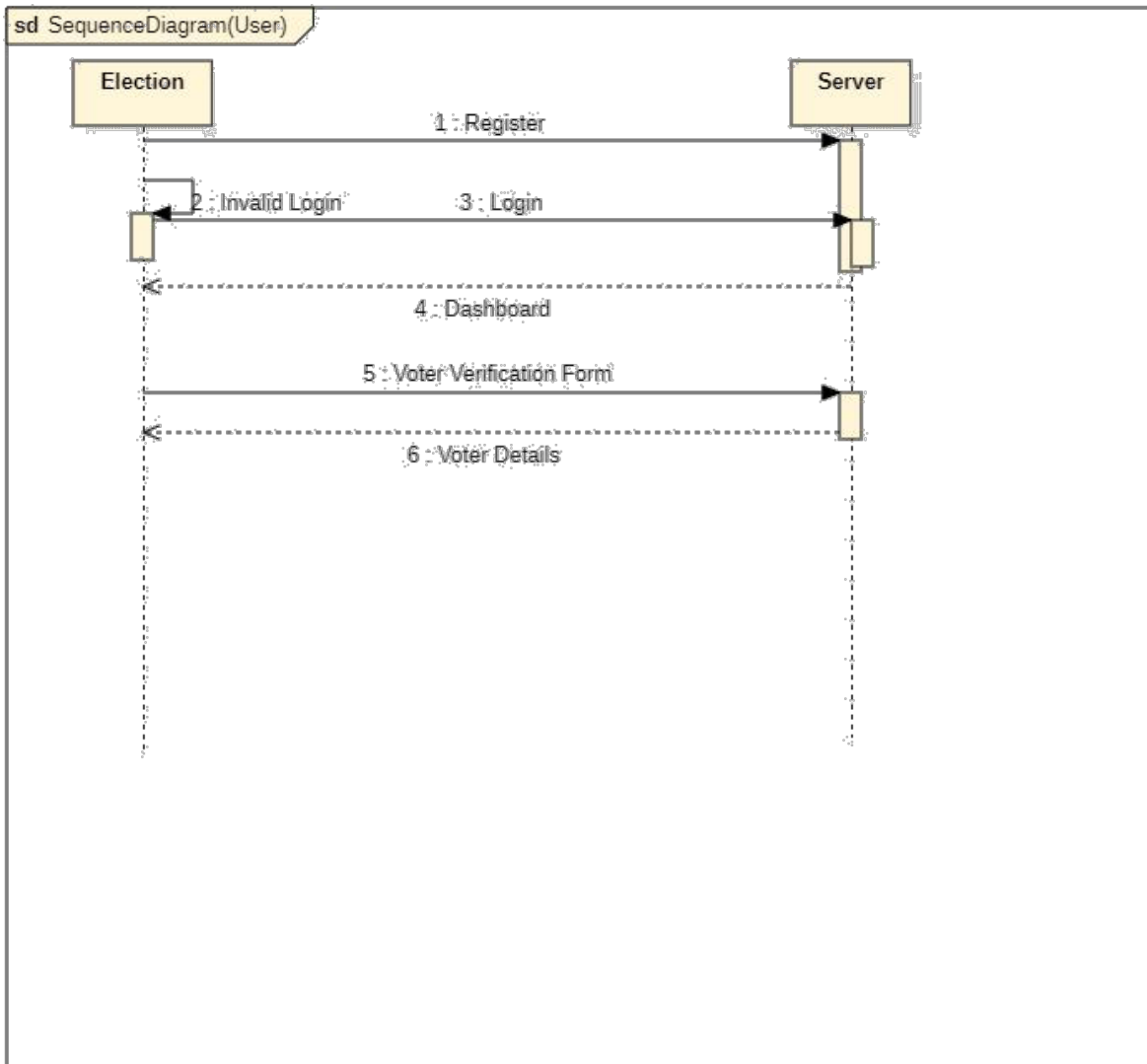
In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.



6.2.3 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

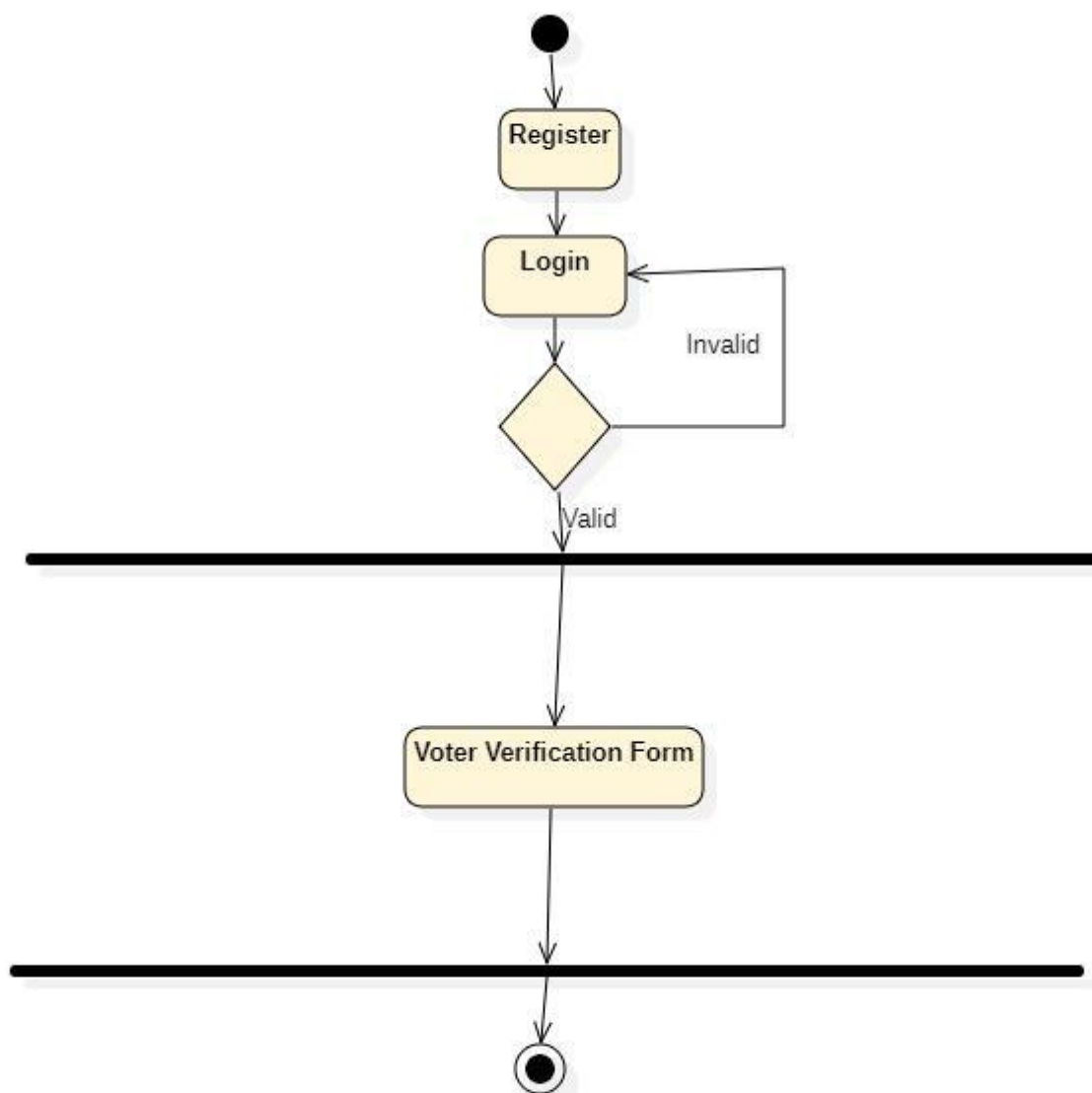




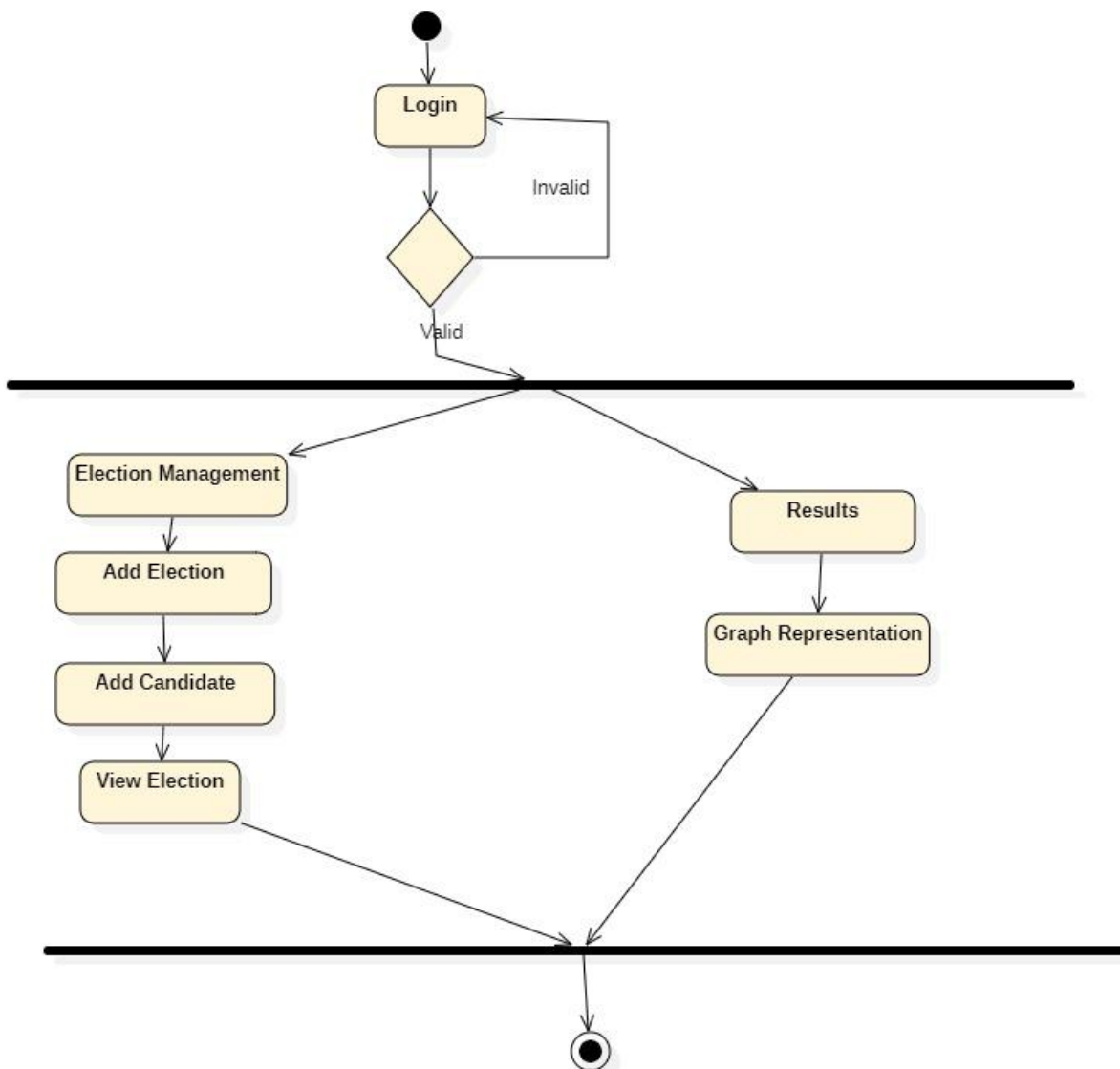
6.2.4 ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

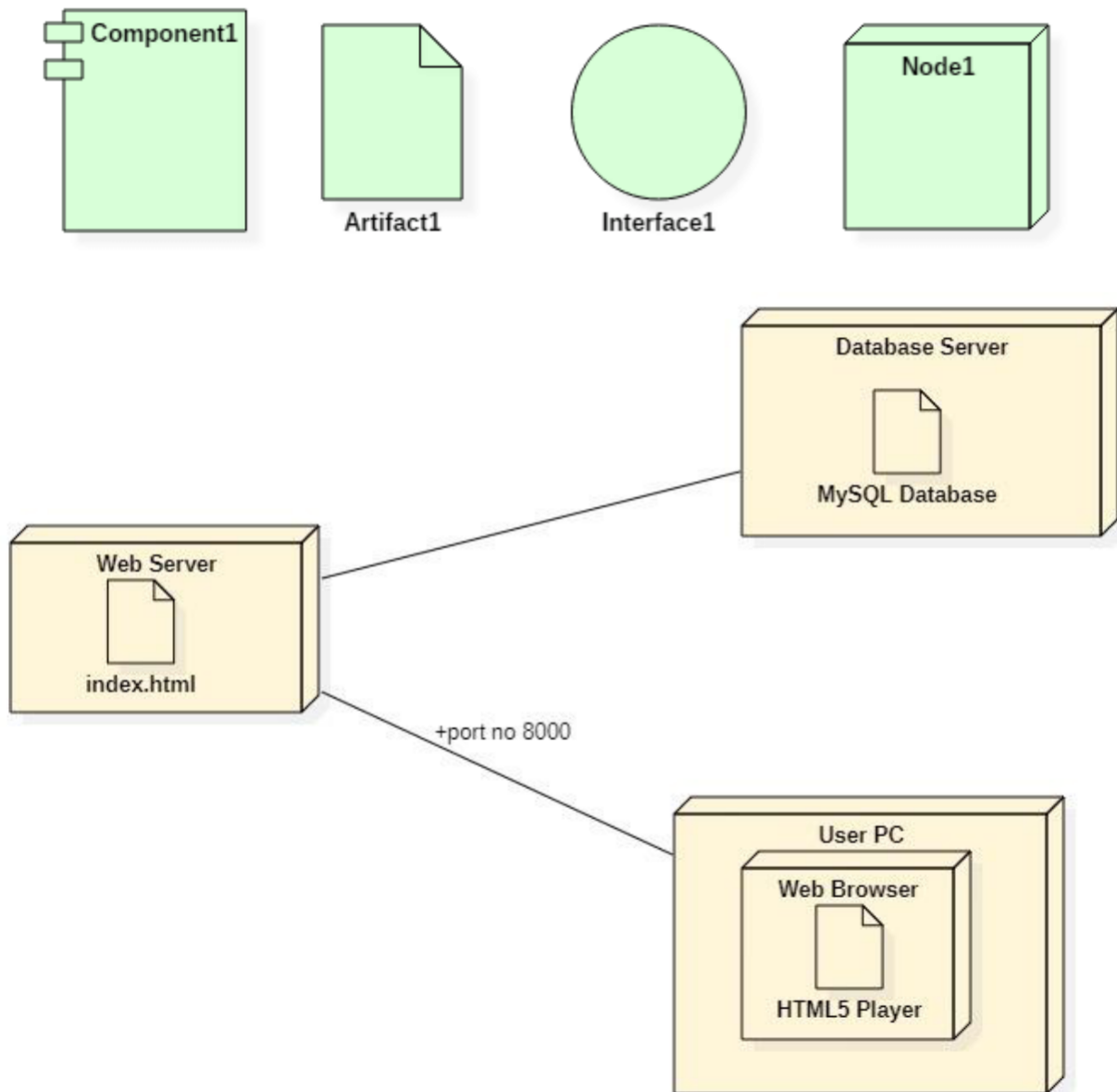
USER:



ADMIN:



6.2.5 DEPLOYMENT DIAGRAM



CHAPTER-7

INPUT AND OUTPUT DESIGN

7.1 INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

7.1.1 OBJECTIVES

1.Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.

2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.

3.When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus, the objective of input design is to create an input layout that is easy to follow.

7.2 OUTPUT DESIGN

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.

2. Select methods for presenting information.

3. Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- Convey information about past activities, current status or projections of the
- Future.
- Signal important events, opportunities, problems, or warnings.
- Trigger an action.
- Confirm an action.

CHAPTER 8

IMPLEMENTATION

8.1 MODULES:

- Election
- Admin

8.2 MODULE DESCRIPTION

User registration and authentication module: This module handles the registration and authentication of voters, verifying their identities using secure methods such as biometrics, encryption, or multi-factor authentication.

Election creation and management module: This module is responsible for creating and managing different elections or voting events, including defining the candidates or options, setting up voting rules, and managing ballot templates.

Voting module: This module handles the actual voting process, allowing registered voters to cast their votes securely and anonymously using their authenticated credentials. It may involve smart contracts on the blockchain to ensure transparency and integrity of the votes.

Vote counting and result verification module: This module is responsible for counting the votes and verifying the results, ensuring accuracy and integrity. It may involve cryptographic techniques to protect the votes from tampering or manipulation.

Blockchain integration module: This module is responsible for integrating the online voting system with a blockchain platform, such as Ethereum or Hyperledger, for decentralized and transparent record-keeping of the votes and results.

Security and privacy module: This module ensures the security and privacy of the online voting system, including measures such as encryption, access control, and audit trails to prevent unauthorized access or data breaches.

Administration and management module: This module handles the administration and management of the online voting system, including user management, ballot configuration, monitoring, and reporting. It may involve a user-friendly interface for administrators to manage the system efficiently.

CHAPTER 9

SOFTWARE ENVIRONMENT

9.1 What is Python :

Below are some facts about Python.

Python is currently the most widely used multi-purpose, high-level programming language.

Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java.

Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.

Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber... etc.

The biggest strength of Python is huge collection of standard library which can be used for the following –

- Machine Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like Opencv, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia

Advantages of Python: -

Let's see how Python dominates over other languages.

1. Extensive Libraries

Python downloads with an extensive library and it *contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more*. So, we don't have to write the complete code for that manually.

2. Extensible

As we have seen earlier, Python can be **extended to other languages**. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add **scripting capabilities** to our code in the other language.

4. Improved Productivity

The language's simplicity and extensive libraries render programmers **more productive** than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world.

When working with Java, you may have to create a class to print '**Hello World**'. But in Python, just a print statement will do. It is also quite **easy to learn, understand, and code**. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and **indentation is mandatory**. This further aids the readability of the code.

8. Object-Oriented

This language supports both the **procedural and object-oriented** programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the **encapsulation of data** and functions into one.

9. Free and Open-Source

Like we said earlier, Python is **freely available**. But not only can you **download Python** for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to **code only once**, and you can run it anywhere. This is called **Write Once Run Anywhere (WORA)**. However, you need to be careful enough not to include any system-dependent features.

11. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, **debugging is easier** than in compiled languages.

Any doubts till now in the advantages of Python? Mention in the comment section.

Advantages of Python Over Other Languages :

1. Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform data analysis and **machine learning**, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in **slow execution**. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the **client-side**. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called **Carbonnelle**.

The reason it is not so famous despite the existence of Brython is that it isn't that secure.

3. Design Restrictions

As you know, Python is **dynamically-typed**. This means that you don't need to declare the type of variable while writing the code. It uses **duck-typing**. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can **raise run-time errors**.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like **JDBC (Java DataBase Connectivity)** and **ODBC (Open DataBase Connectivity)**, Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

This was all about the Advantages and Disadvantages of Python Programming Language.

History of Python : -

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general-purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde & Informatica). The greatest achievement of ABC was to influence the design of Python. Python was conceptualized in the late 1980s. Guido van Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating system. In an interview with Bill Venner¹, Guido van Rossum said: "In the early 1980s, I worked as an implementer on a team building a language called ABC at Centrum voor Wiskunde en Informatica (CWI). I don't know how well people know ABC's influence on Python. I try to mention ABC's influence because I'm indebted to everything I learned during that project and to the people who worked on it." Later on in the same interview, Guido van Rossum continued: "I remembered all my experience and some of my frustration with ABC. I decided to try to design a simple scripting language that possessed some of ABC's better properties, but without its problems. So I started typing. I created a simple virtual machine, a simple parser, and a simple runtime. I made my own version of the various ABC parts that I liked. I created a basic syntax, used indentation for statement grouping instead of curly braces or begin-end blocks, and developed a small number of powerful data types: a hash table (or dictionary, as we call it), a list, strings, and numbers."

What is Machine Learning : -

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't. Machine learning is often categorized as a subfield of artificial intelligence, but I find that categorization can often be misleading at first brush. The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of *building models of data*.

Fundamentally, machine learning involves building mathematical models to help understand data. "Learning" enters the fray when we give these models *tunable parameters* that can be adapted to observed data; in this way the program can be considered to be "learning" from the data. Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data. I'll leave to the reader the more philosophical digression regarding the extent

to which this type of mathematical, model-based "learning" is similar to the "learning" exhibited by the human brain. Understanding the problem setting in machine learning is essential to using these tools effectively, and so we will start with some broad categorizations of the types of approaches we'll discuss here.

Categories Of Machine Learning :-

At the most fundamental level, machine learning can be categorized into two main types: supervised learning and unsupervised learning.

Supervised learning involves somehow modeling the relationship between measured features of data and some label associated with the data; once this model is determined, it can be used to apply labels to new, unknown data. This is further subdivided into *classification* tasks and *regression* tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities. We will see examples of both types of supervised learning in the following section.

Unsupervised learning involves modeling the features of a dataset without reference to any label, and is often described as "letting the dataset speak for itself." These models include tasks such as *clustering* and *dimensionality reduction*. Clustering algorithms identify distinct groups of data, while dimensionality reduction algorithms search for more succinct representations of the data. We will see examples of both types of unsupervised learning in the following section.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and haven't surpassed human intelligence in many aspects. Then the question is that what is the need to make machine learn? The most suitable reason for doing this is, "to make decisions, based on data, with efficiency and scale".

Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programming logic, in the problems that cannot be programmed inherently. The fact is that we can't do without human

intelligence, but other aspect is that we all need to solve real-world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning :-

While Machine Learning is rapidly evolving, making significant strides with cybersecurity and autonomous cars, this segment of AI as whole still has a long way to go. The reason behind is that ML has not been able to overcome number of challenges. The challenges that ML is facing currently are –

Quality of data – Having good-quality data for ML algorithms is one of the biggest challenges. Use of low-quality data leads to the problems related to data preprocessing and feature extraction.

Time-Consuming task – Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and retrieval.

Lack of specialist persons – As ML technology is still in its infancy stage, availability of expert resources is a tough job.

No clear objective for formulating business problems – Having no clear objective and well-defined goal for business problems is another key challenge for ML because this technology is not that mature yet.

Issue of overfitting & underfitting – If the model is overfitting or underfitting, it cannot be represented well for the problem.

Curse of dimensionality – Another challenge ML model faces is too many features of data points. This can be a real hindrance.

Difficulty in deployment – Complexity of the ML model makes it quite difficult to be deployed in real life.

Applications of Machines Learning :-

Machine Learning is the most rapidly growing technology and according to researchers we are in the golden year of AI and ML. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML-

- Emotion analysis
- Sentiment analysis

- Error detection and prevention
- Weather forecasting and prediction
- Stock market analysis and forecasting
- Speech synthesis
- Speech recognition
- Customer segmentation
- Object recognition
- Fraud detection
- Fraud prevention
- Recommendation of products to customer in online shopping

How to Start Learning Machine Learning?

Arthur Samuel coined the term “**Machine Learning**” in 1959 and defined it as a “**Field of study that gives computers the capability to learn without being explicitly programmed**”.

And that was the beginning of Machine Learning! In modern times, Machine Learning is one of the most popular (if not the most!) career choices. According to Indeed, Machine Learning Engineer Is The Best Job of 2019 with a 344% growth and an average base salary of **\$146,085** per year.

But there is still a lot of doubt about what exactly is Machine Learning and how to start learning it? So this article deals with the Basics of Machine Learning and also the path you can follow to eventually become a full-fledged Machine Learning Engineer. Now let's get started!!!

How to start learning ML?

This is a rough roadmap you can follow on your way to becoming an insanely talented Machine Learning Engineer. Of course, you can always modify the steps according to your needs to reach your desired end-goal!

Step 1 – Understand the Prerequisites

In case you are a genius, you could start ML directly but normally, there are some prerequisites that you need to know which include Linear Algebra, Multivariate Calculus, Statistics, and Python. And

if you don't know these, never fear! You don't need a Ph.D. degree in these topics to get started but you do need a basic understanding.

(a) Learn Linear Algebra and Multivariate Calculus

Both Linear Algebra and Multivariate Calculus are important in Machine Learning. However, the extent to which you need them depends on your role as a data scientist. If you are more focused on application heavy machine learning, then you will not be that heavily focused on maths as there are many common libraries available. But if you want to focus on R&D in Machine Learning, then mastery of Linear Algebra and Multivariate Calculus is very important as you will have to implement many ML algorithms from scratch.

(b) Learn Statistics

Data plays a huge role in Machine Learning. In fact, around 80% of your time as an ML expert will be spent collecting and cleaning data. And statistics is a field that handles the collection, analysis, and presentation of data. So it is no surprise that you need to learn it!!! Some of the key concepts in statistics that are important are Statistical Significance, Probability Distributions, Hypothesis Testing, Regression, etc. Also, Bayesian Thinking is also a very important part of ML which deals with various concepts like Conditional Probability, Priors, and Posteriors, Maximum Likelihood, etc.

(c) Learn Python

Some people prefer to skip Linear Algebra, Multivariate Calculus and Statistics and learn them as they go along with trial and error. But the one thing that you absolutely cannot skip is Python! While there are other languages you can use for Machine Learning like R, Scala, etc. Python is currently the most popular language for ML. In fact, there are many Python libraries that are specifically useful for Artificial Intelligence and Machine Learning such as Keras, TensorFlow, Scikit-learn, etc.

So if you want to learn ML, it's best if you learn Python! You can do that using various online resources and courses such as **Fork Python** available Free on GeeksforGeeks.

Step 2 – Learn Various ML Concepts

Now that you are done with the prerequisites, you can move on to actually learning ML (Which is the fun part!!!) It's best to start with the basics and then move on to the more complicated stuff. Some of the basic concepts in ML are:

(a) Terminologies of Machine Learning

- **Model** – A model is a specific representation learned from data by applying some machine learning algorithm. A model is also called a hypothesis.
- **Feature** – A feature is an individual measurable property of the data. A set of numeric features can be conveniently described by a feature vector. Feature vectors are fed as input to the model. For example, in order to predict a fruit, there may be features like color, smell, taste, etc.
- **Target (Label)** – A target variable or label is the value to be predicted by our model. For the fruit example discussed in the feature section, the label with each set of input would be the name of the fruit like apple, orange, banana, etc.
- **Training** – The idea is to give a set of inputs(features) and it's expected outputs(labels), so after training, we will have a model (hypothesis) that will then map new data to one of the categories trained on.
- **Prediction** – Once our model is ready, it can be fed a set of inputs to which it will provide a predicted output(label).

(b) Types of Machine Learning

- **Supervised Learning** – This involves learning from a training dataset with labeled data using classification and regression models. This learning process continues until the required level of performance is achieved.
- **Unsupervised Learning** – This involves using unlabelled data and then finding the underlying structure in the data in order to learn more and more about the data itself using factor and cluster analysis models.
- **Semi-supervised Learning** – This involves using unlabelled data like Unsupervised Learning with a small amount of labeled data. Using labeled data vastly increases the learning accuracy and is also more cost-effective than Supervised Learning.

- **Reinforcement Learning** – This involves learning optimal actions through trial and error. So the next action is decided by learning behaviors that are based on the current state and that will maximize the reward in the future.

Advantages of Machine learning :-

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation)

With ML, you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus softwares; they learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement

As **ML algorithms** gain experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say you need to make a weather forecast model. As the amount of data you have keeps growing, your algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data

Machine Learning algorithms are good at handling data that are multi-dimensional and multi-variety, and they can do this in dynamic or uncertain environments.

5. Wide Applications

You could be an e-tailer or a healthcare provider and make ML work for you. Where it does apply, it holds the capability to help deliver a much more personal experience to customers while also targeting the right customers.

Disadvantages of Machine Learning :-

1. Data Acquisition

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for you.

3. Interpretation of Results

Another major challenge is the ability to accurately interpret results generated by the algorithms. You must also carefully choose the algorithms for your purpose.

4. High error-susceptibility

Machine Learning is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time. And when they do get noticed, it takes quite some time to recognize the source of the issue, and even longer to correct it.

Python Development Steps : -

Guido Van Rossum published the first version of Python code (version 0.9.0) at alt.sources in February 1991. This release included already exception handling, functions, and the core data types of list, dict, str and others. It was also object oriented and had a module system. Python version 1.0 was released in January 1994. The major new features included in this release were the functional programming tools lambda, map, filter and reduce, which Guido Van Rossum never liked. Six and a half years later in October 2000, Python 2.0 was introduced. This release included list comprehensions, a full garbage collector and it was supporting unicode. Python flourished for another 8 years in the versions 2.x before the next major release as Python 3.0 (also known as "Python 3000" and "Py3K") was released. Python 3 is not backwards compatible with Python 2.x. The emphasis in

Python 3 had been on the removal of duplicate programming constructs and modules, thus fulfilling or coming close to fulfilling the 13th law of the Zen of Python: "There should be one -- and preferably only one -- obvious way to do it." Some changes in Python 7.3:

- Print is now a function
- Views and iterators instead of lists
- The rules for ordering comparisons have been simplified. E.g. a heterogeneous list cannot be sorted, because all the elements of a list must be comparable to each other.
- There is only one integer type left, i.e. int. long is int as well.
- The division of two integers returns a float instead of an integer. "/" can be used to have the "old" behaviour.
- Text Vs. Data Instead Of Unicode Vs. 8-bit

Purpose :-

We demonstrated that our approach enables successful segmentation of intra-retinal layers—even with low-quality images containing speckle noise, low contrast, and different intensity ranges throughout—with the assistance of the ANIS feature.

Python

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed

of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Modules Used in Project :-

Tensorflow

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks. It is used for both research and production at Google.

TensorFlow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0 open-source license on November 9, 2015.

Numpy

Numpy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, Numpy can also be used as an efficient multi-dimensional container of generic data. Arbitrary data-types can be defined using Numpy which allows Numpy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a

wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and IPython shells, the Jupyter Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and thumbnail gallery.

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object oriented interface or via a set of functions familiar to MATLAB users.

Scikit – learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use. **Python**

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skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Install Python Step-by-Step in Windows and Mac :

Python a versatile programming language doesn't come pre-installed on your computer devices. Python was first released in the year 1991 and until today it is a very popular high-level programming language. Its style philosophy emphasizes code readability with its notable use of great whitespace.

The object-oriented approach and language construct provided by Python enables programmers to write both clear and logical code for projects. This software does not come pre-packaged with Windows.

How to Install Python on Windows and Mac :

There have been several updates in the Python version over the years. The question is how to install Python? It might be confusing for the beginner who is willing to start learning Python but this tutorial will solve your query. The latest or the newest version of Python is version 3.7.4 or in other words, it is Python 3.

Note: The python version 3.7.4 cannot be used on Windows XP or earlier devices.

Before you start with the installation process of Python. First, you need to know about your **System Requirements**. Based on your system type i.e. operating system and based processor, you must download the python version. My system type is a **Windows 64-bit operating system**. So the steps below are to install python version 3.7.4 on Windows 7 device or to install Python 3. [Download the Python Cheatsheet here.](#) The steps on how to install Python on Windows 10, 8 and 7 are **divided into 4 parts** to help understand better.

Download the Correct version into the system

Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link: <https://www.python.org>



Now, check for the latest and the correct version for your operating system.

Step 2: Click on the Download Tab.



Step 3: You can either select the Download Python for windows 3.7.4 button in Yellow Color or you can scroll further down and click on download with respective to their version. Here, we are downloading the most recent python version for windows 3.7.4

Looking for a specific release?

Python releases by version number

Release version	Release date		Click for more
Python 3.7.4	July 9, 2019	Download	Release Notes
Python 3.6.9	July 1, 2019	Download	Release Notes
Python 3.7.3	March 25, 2019	Download	Release Notes
Python 3.6.8	March 19, 2019	Download	Release Notes
Python 3.7.2	March 22, 2018	Download	Release Notes
Python 3.6.7	March 4, 2018	Download	Release Notes
Python 3.7.1	Dec 22, 2018	Download	Release Notes

Step 4: Scroll down the page until you find the Files option.

Step 5: Here you see a different version of python along with the operating system.

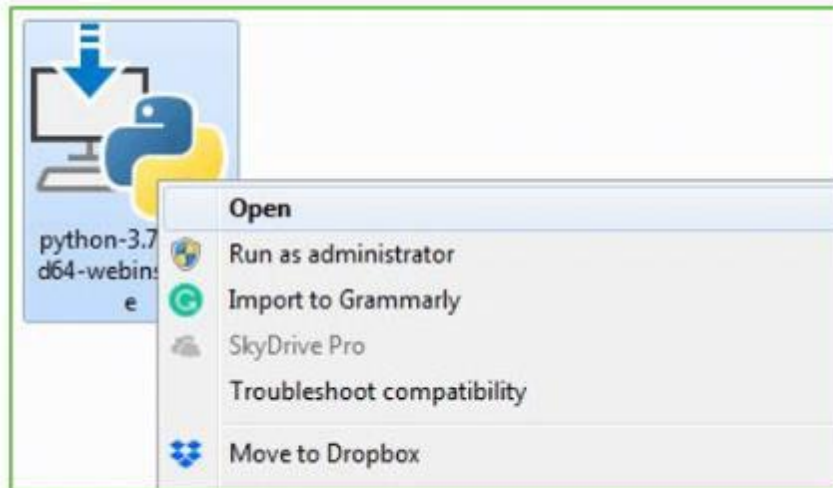
Files					
Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		68111671e5b2db4ae77b9ab01b709be	23017663	SG
XZ compressed source tarball	Source release		d33e4aae6097051c2mca45ee3604803	17131432	SG
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.6 and later	6428b4fa7583daf1a442cbalce08e6	34898436	SG
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	5dd605c38217a45773bf5e4a936b241f	20082845	SG
Windows help file	Windows		063999573a2c96b2ac5fcaade6b4f7cd2	8131761	SG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	9b093cfd29c0f8abe8318aa0729a2	7504391	SG
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	a702b4b0af76d9db3543a383e563400	26680368	SG
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	28cb1c6088bd72aebf53a3bd351b4bd2	1362904	SG
Windows x86 embeddable zip file	Windows		9fab1bd18841879fda94132574139d8	6741626	SG
Windows x86 executable installer	Windows		33cc02942a5446a3d8451476394789	25663848	SG
Windows x86 web-based installer	Windows		1b670cfa5d317df82c30983ea371087c	1324608	SG

- To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 web-based installer.
- To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows x86-64 web-based installer. Here we will install Windows x86-64 web-based installer. Here your first part regarding which version of python is to be downloaded is completed. Now we move ahead with the second part in installing python i.e. Installation

Note: To know the changes or updates that are made in the version you can click on the Release Note Option.

Installation of Python

Step 1: Go to Download and Open the downloaded python version to carry out the installation process.



Step 2: Before you click on Install Now, Make sure to put a tick on Add Python 3.7 to PATH.



Step 3: Click on Install NOW After the installation is successful. Click on Close.



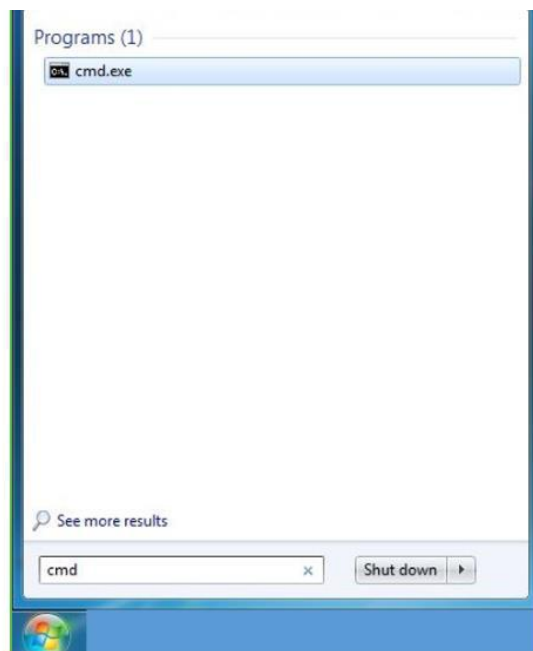
With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

Note: The installation process might take a couple of minutes.

Verify the Python Installation

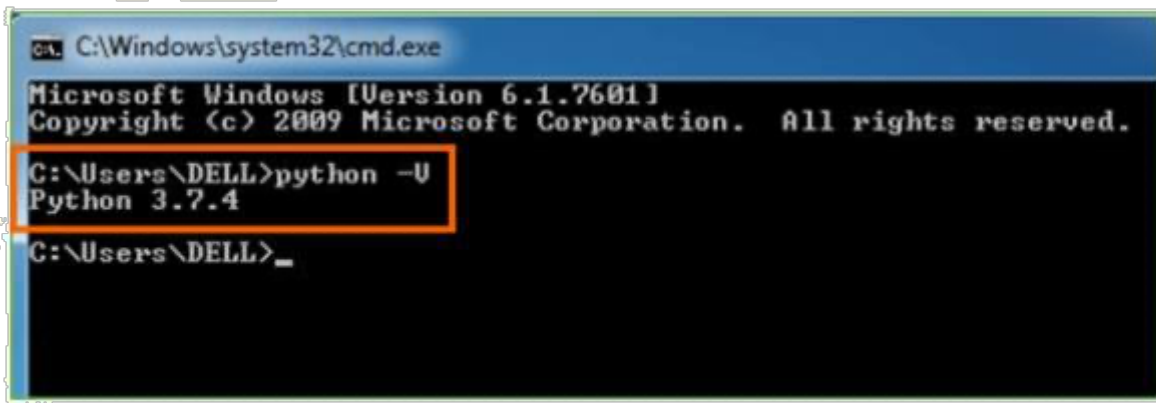
Step 1: Click on Start

Step 2: In the Windows Run Command, type “cmd”.



Step 3: Open the Command prompt option.

Step 4: Let us test whether the python is correctly installed. Type **python -V** and press Enter.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\DELL>python -U
Python 3.7.4

C:\Users\DELL>_
```

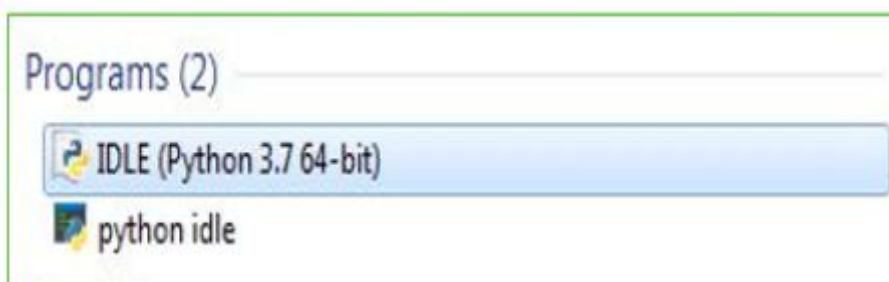
Step 5: You will get the answer as 3.7.4

Note: If you have any of the earlier versions of Python already installed. You must first uninstall the earlier version and then install the new one.

Check how the Python IDLE works

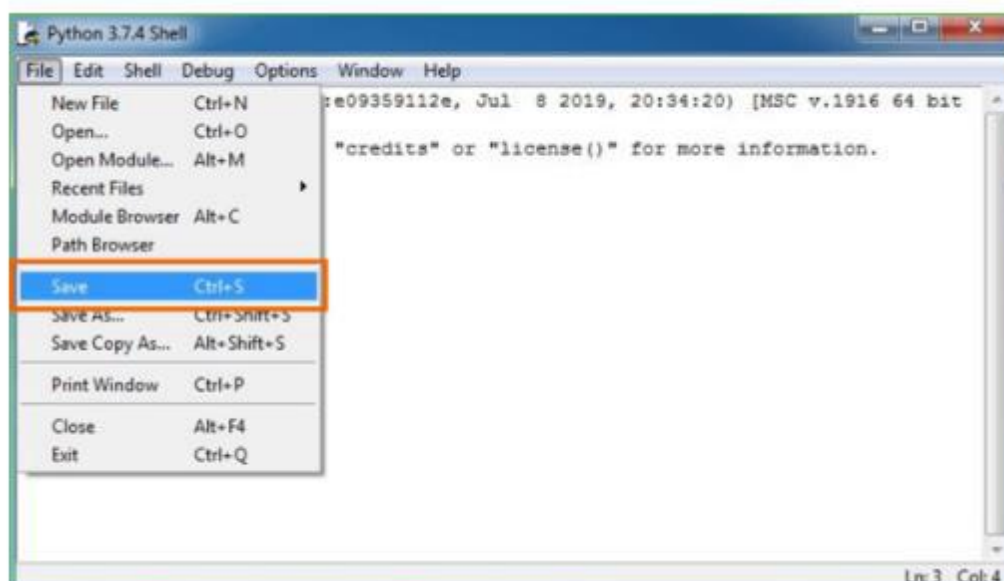
Step 1: Click on Start

Step 2: In the Windows Run command, type “python idle”.



Step 3: Click on IDLE (Python 3.7 64-bit) and launch the program

Step 4: To go ahead with working in IDLE you must first save the file. **Click on File > Click on Save**



Step 5: Name the file and save as type should be Python files. Click on SAVE. Here I have named the files as Hey World.

Step 6: Now for e.g. **enter print**

9.2 SOURCE CODE

Manage.py

```
#!/usr/bin/env python
"""Django's command-line utility for administrative tasks."""
import os
import sys

def main():
    """Run administrative tasks."""
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'decentralizedvoting.settings')
    try:
        from django.core.management import
execute_from_command_line except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
            "forget to activate a virtual environment?"
        ) from exc
execute_from_command_line(sys.argv)

if __name__ == '__main__':
    main()
```

index.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">
```

```

<head>
  <title>Smart Electronic voting
  Application</title> <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">

  <link href="https://fonts.googleapis.com/css?family=Muli:300,400,700,900"
  rel="stylesheet"> <link href="{% static 'main/images/logo.png' %}" rel="icon">

  <link rel="stylesheet" href="{% static 'main/fonts/icomoon/style.css' %}">

  <link rel="stylesheet" href="{% static 'main/css/bootstrap.min.css' %}">
  <link rel="stylesheet" href="{% static 'main/css/jquery-ui.css' %}">
  <link rel="stylesheet" href="{% static 'main/css/owl.carousel.min.css' %}"> <link
  rel="stylesheet" href="{% static 'main/css/owl.theme.default.min.css' %}"> <link
  rel="stylesheet" href="{% static 'main/css/owl.theme.default.min.css' %}">

  <link rel="stylesheet" href="{% static 'main/css/jquery.fancybox.min.css' %}">

  <link rel="stylesheet" href="{% static 'main/css/bootstrap-datepicker.css' %}">

  <link rel="stylesheet" href="{% static 'main/fonts/flaticon/font/flaticon.css' %}">

  <link rel="stylesheet" href="{% static 'main/css/aos.css' %}">

  <link rel="stylesheet" href="{% static 'main/css/style.css' %}">
  <script src="{% static 'main/js/sweetalert.min.js' %}"></script>

</head>

<body data-spy="scroll" data-target=".site-navbar-target" data-
offset="300"> <div class="site-wrap">
  <div class="site-mobile-menu site-navbar-target">

```

```

<div class="site-mobile-menu-header">
  <div class="site-mobile-menu-close mt-3">
    <span class="icon-close2 js-menu-
toggle"></span> </div>
  </div>
  <div class="site-mobile-menu-body"></div>
</div>

<!-- header start -->

<header class="site-navbar navv py-4 js-sticky-header site-navbar-target"
role="banner"> <div class="container-fluid">
  <div class="d-flex align-items-center">
    <div class="site-logo mr-auto">

      <a href="{% url 'index' %}"> Blockchain Voting</a>
    </div>
    <div class="mx-auto text-center">
      <nav class="site-navigation position-relative text-right" role="navigation">
        <ul class="site-menu main-menu js-clone-nav mx-auto d-none d-lg-block m-0 p-
0"> <li><a href="{% url 'index' %}" class="nav-link active">Home</a></li>
        <li><a href="{% url 'voter_elections' %}" class="nav-link">Elections</a></li>
        <li><a href="{% url 'voter_results' %}" class="nav-link">Results</a></li>
        <li><a href="{% url 'admin_login' %}" class="nav-link">Admin</a></li>
        <li><a href="{% url 'contact' %}" class="nav-link">Contact</a></li>
        </ul>
      </nav>
    </div>
    <div class="d-inline-block d-lg-none site-menu-toggle js-menu-toggle float-right
ml-auto">

      <a href="#" class="" style="color: black;"><span class="icon-menu
h3"></span></a> </div>

```

```

</div>
</div>
</header>
<!-- header end -->

<!-- section start -->

<div class="intro-section" id="home-section">
  <div class="slide-1" style="background-image: url('/static/main/images/eci.jpeg');height:
140vh;"
    data-stellar-background-ratio="0.7">
    <div class="container pt-5">
      <div class="container pt-5">
        <div class="row align-items-center">
          <div class="col-12">
            <div class="row align-items-center">
              <div class="col-12 mb-4 text-center mx-auto">
                <h1 data-aos="fade-up" data-aos-delay="100">DECENTRALIZED VOTING</h1>
                <p class="mb-4" data-aos="fade-up" data-aos-delay="200">Voting is one of the
processes that
                allow citizens to find their place in society, as well as one
                of the rights to elect a worthy and modest leader. There are
                a lot of voting systems that aren't secure, therefore by
                incorporating the blockchain, security is ensured.
                Verification of Aadhar is
                used to create a digital signature has a significant role
                to play. </p>
            </div>
            <div class="col-lg-6 mb-4 text-center mx-auto mt-5">
              <h1 data-aos="fade-up" data-aos-delay="100">VOTE FOR CHANGE
              </h1> <p class="mb-4" data-aos="fade-up" data-aos-delay="200"> <span
class="h3">"</span> The vote is
                precious. It is the most powerful non-violent tool we have in a democratic society,
                and we must

```

```

        use it.<span class="h3">"</span></p>
    </div>
    <div class="col-12 mb-4 text-center mx-auto mt-5">
        <h1 data-aos="fade-up" data-aos-delay="100">Created By :</h1>
        <p class="mb-4 text-light" data-aos="fade-up" data-aos-
delay="200"><a href="https://codebook.in/"
        class="text-light"
        target="_blank">Codebook.in</a></p> </div>
    </div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- section end-->

```

```

{% if messages %}
{% for message in messages %}
{% if message.level == DEFAULT_MESSAGE_LEVELS.ERROR
%} <script>
    swal({
        title: "Error :(",
        text: "{{message}}",
        icon: "error",
        button: "OK",
    });
</script>
{% elif message.level == DEFAULT_MESSAGE_LEVELS.SUCCESS
%} <script>
    swal({
        title: "Success :)",
        text: "{{message}}",
        icon: "success",

```

```
        button: "OK",
    });
</script>
```

```
{% elif message.level == DEFAULT_MESSAGE_LEVELS.INFO
%} <script>
    swal({
        title: "Info :(",
        text: "{{message}}",
        icon: "info",
        button: "OK",
    });
</script>
```

```
{% elif message.level == DEFAULT_MESSAGE_LEVELS.WARNING
%} <script>
    swal({
        title: "Warning :(",
        text: "{{message}}",
        icon: "warning",
        button: "OK",
    });
</script>
```

```
{% endif %}
{% endfor %}
{% endif %}
<!-- footer start -->
```

```
<footer class="footer-section foot bg-white">
    <hr>
```

```
    <div class="container">
        <div class="row">
```

```
<div class="col-md-4">
  <h3>About codebook</h3>
  <p>With an impressive list of highly qualified employees, our company is one of
the most successful startups
  in Hyderabad.
```

```

  There is a university of code called Codebook where you can search for any project
title and download it
```

```
    without any hassle.</p>
</div>
```

```
<div class="col-md-3 ml-auto">
```

```
  <h3>Links</h3>
```

```
  <ul class="list-unstyled footer-links">
```

```
    <li><a href="{% url 'index' %}">Home</a></li>
```

```

    <li><a href="{% url 'admin_login' %}">Admin</a></li>
```

```
    <li><a href="{% url 'contact' %}">Contact Us</a></li>
```

```
  </ul>
```

```
</div>
```

```
<p class="text-center">
```

```
  Copyright All rights reserved | This project is made with <i class="icon-heart"
aria-hidden="true"></i> by
```

```
  <a href="https://codebook.in/"
```

```
target="codebook.in">Codebook</a> </p>
```

```
</div>
```

```
</div>
```

```
</footer>
```

```
<!-- footer end -->
```

```
</div>
```

```

<script src="{% static 'main/js/jquery-3.3.1.min.js' %}"></script>
<script src="{% static 'main/js/jquery-migrate-3.0.1.min.js' %}"></script>
<script src="{% static 'main/js/jquery-ui.js' %}"></script>
<script src="{% static 'main/js/popper.min.js' %}"></script> <script
src="{% static 'main/js/bootstrap.min.js' %}"></script> <script src="{%
static 'main/js/owl.carousel.min.js' %}"></script> <script src="{% static
'main/js/jquery.stellar.min.js' %}"></script> <script src="{% static
'main/js/jquery.countdown.min.js' %}"></script> <script src="{% static
'main/js/bootstrap-datepicker.min.js' %}"></script> <script src="{%
static 'main/js/jquery.easing.1.3.js' %}"></script> <script src="{% static
'main/js/aos.js' %}"></script>
<script src="{% static 'main/js/jquery.fancybox.min.js' %}"></script>
<script src="{% static 'main/js/jquery.sticky.js' %}"></script>

```

```

<script src="{% static 'main/js/main.js' %}"></script>

```

```

</body>

```

```

</html>

```

ADMIN-LOGIN.HTML

```

{% load static %}

```

```

<!DOCTYPE html>

```

```

<html lang="en">

```

```

<head>

```

```

<meta charset="utf-8">

```

```

<meta http-equiv="X-UA-Compatible" content="IE=edge">

```

```

<meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">

```

```

<meta name="description" content="">

```

```

<meta name="author" content="">

```

```

<link href="{% static 'main/images/logo.png' %}"
rel="icon"> <title>RuangAdmin - Login</title>
<link href="{% static 'admin/vendor/fontawesome-free/css/all.min.css' %}"
rel="stylesheet" type="text/css">
<link href="{% static 'admin/vendor/bootstrap/css/bootstrap.min.css' %}"
rel="stylesheet" type="text/css">
<link href="{% static 'admin/css/ruang-admin.min.css' %}" rel="stylesheet">

<link href="https://fonts.googleapis.com/css?family=Muli:300,400,700,900" rel="stylesheet">
<link rel="stylesheet" href="{% static 'main/fonts/icomoon/style.css' %}">

<link rel="stylesheet" href="{% static 'main/css/bootstrap.min.css' %}">
<link rel="stylesheet" href="{% static 'main/css/jquery-ui.css' %}">
<link rel="stylesheet" href="{% static 'main/css/owl.carousel.min.css' %}"> <link
rel="stylesheet" href="{% static 'main/css/owl.theme.default.min.css' %}"> <link
rel="stylesheet" href="{% static 'main/css/owl.theme.default.min.css' %}">

<link rel="stylesheet" href="{% static 'main/css/jquery.fancybox.min.css' %}">

<link rel="stylesheet" href="{% static 'main/css/bootstrap-datepicker.css' %}">

<link rel="stylesheet" href="{% static 'main/fonts/flaticon/font/flaticon.css' %}">

<link rel="stylesheet" href="{% static 'main/css/aos.css' %}">

<link rel="stylesheet" href="{% static 'main/css/style.css'
%}"> <script src="{% static 'main/js/sweetalert.min.js'
%}"></script> <script type="text/javascript" >
    function validation(){
        let username = document.myform.username
        let password = document.myform.password

        if (username.value.length <= 0){
            alert('Please enter username')

```

```

        username.focus()
        return false;
    }
    if (password.value.length <=0){
        alert('Please enter password')
        password.focus()
        return false;
    }
}
</script>

```

```

</head>

```

```

<body class="bg-gradient-login" data-spy="scroll" data-target=".site-navbar-target"
data-offset="300">

```

```

<div class="site-wrap">

```

```

<div class="site-mobile-menu site-navbar-target
"> <div class="site-mobile-menu-header">
    <div class="site-mobile-menu-close mt-3 ">
        <span class="icon-close2 js-menu-toggle"></span>
    </div>
</div>
<div class="site-mobile-menu-body"></div>
</div>

```

```

<header class="site-navbar navv py-4 js-sticky-header site-navbar-target"
role="banner"> <div class="container-fluid">
    <div class="d-flex align-items-center">
        <div class="site-logo mr-auto">

            <a href="{% url 'index' %}"> Blockchain Voting</a>

```

```

</div>
<div class="mx-auto text-center">
  <nav class="site-navigation position-relative text-right" role="navigation">
    <ul class="site-menu main-menu js-clone-nav mx-auto d-none d-lg-block m-0 p-0">
      <li><a href="{% url 'index' %}" class="nav-link ">Home</a></li>
      <li><a href="{% url 'voter_elections' %}" class="nav-link">Elections</a></li>
      <li><a href="{% url 'voter_results' %}" class="nav-link">Results</a></li>
      <li><a href="{% url 'admin_login' %}" class="nav-link active">Admin</a></li>
      <li><a href="{% url 'contact' %}" class="nav-link">Contact</a></li>
    </ul>
  </nav>
</div>
<div class="d-inline-block d-lg-none site-menu-toggle js-menu-toggle float-right ml-auto">

  <a href="#" class=""><span class="icon-menu h3"></span></a> </div>
</div>
</div>
</header>

<div class="intro-section" id="home-section">

  <div class="slide-1" style="background-image: url('/static/main/images/eci.jpeg');"
  data-stellar-background-ratio="0.5">
    <div class="container-login">

      <div class="container ">
        <div class="row justify-content-center pt-5">
          <div class="col-xl-6 col-lg-12 col-md-9 mt-5">
            <div class="card shadow-sm mt-5">
              <div class="card-body p-0">
                <div class="row">
                  <div class="col-lg-12">

```

```

<div class="login-form ">
  <div class="text-center">
    <h1 class="h4 text-gray-900 mb-3">Admin
    Login</h1> </div>
    <form class="user" action="" method="post" name="myform" onsubmit="return
validation()">
      {% csrf_token %}
      <div class="form-group">
        <input type="text" class="form-control" name="username"
id="exampleInputEmail" aria-describedby="emailHelp"
        placeholder="Enter Your Username">
      </div>
      <div class="form-group">
        <input type="password" name="password" class="form-
control" id="exampleInputPassword" placeholder="Password">
      </div>
      <div class="form-group">
        <div class="custom-control custom-checkbox small" style="line-height:
1.5rem;">
          <input type="checkbox" class="custom-control-input" id="customCheck">
          <label class="custom-control-label" for="customCheck">Remember
            Me</label>
        </div>
      </div>
      <div class="form-group">
        <button type="submit" class="btn btn-primary btn-block w-50
mx-auto">Login</button>
      </div>
      <hr>
    </form>
  </div>
</div>
</div>
</div>

```

```

        </div>
    </div>
</div>
</div>

</div>
</div>
</div>

{% if messages %}
{% for message in messages %}
{% if message.level == DEFAULT_MESSAGE_LEVELS.ERROR
%} <script>
    swal({
        title: "Error :(",
        text: "{{message}}",
        icon: "error",
        button: "OK",
    });
</script>

{% elif message.level == DEFAULT_MESSAGE_LEVELS.SUCCESS
%} <script>
    swal({
        title: "Success :)",
        text: "{{message}}",
        icon: "success",
        button: "OK",
    });
</script>

{% elif message.level == DEFAULT_MESSAGE_LEVELS.INFO
%} <script>
    swal({
        title: "Info :(",

```

```

        text: "{{message}}",
        icon: "info",
        button: "OK",
    });
</script>

```

```

{% elif message.level == DEFAULT_MESSAGE_LEVELS.WARNING
%} <script>
    swal({
        title: "Warning :",
        text: "{{message}}",
        icon: "warning",
        button: "OK",
    });
</script>

```

```

{% endif %}
{% endfor %}
{% endif %}

```

```

<footer class="footer-section bg-white">

```

```

    <hr>

```

```

    <div class="container">

```

```

        <div class="row">

```

```

            <div class="col-md-4">

```

```

                <h3>About codebook</h3>

```

```

                <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit. Porro consectetur ut hic
ipsum et veritatis corrupti. Itaque eius soluta optio dolorum temporibus in, atque, quos fugit
sunt sit quaerat dicta.</p>

```

```

            </div>

```

```

        <div class="col-md-3 ml-auto">

```

```

            <h3>Links</h3>

```

```

            <ul class="list-unstyled footer-links">

```

```

        <li><a href="/voter/{% url 'index' %}">Home</a></li>
        <li><a href="{% url 'admin_login' %}">Admin</a></li>
        <li><a href="{% url 'contact' %}">Contact Us</a></li>
    </ul>
</div>

<p class="text-center">
    Copyright All rights reserved | This project is made with <i class="icon-heart" aria-
hidden="true"></i> by <a href="https://codebook.in/" target="codebook.in">Codebook</a>
</p>

</div>
</div>
</footer>
<script src="{% static 'main/js/jquery-3.3.1.min.js' %}"></script>
<script src="{% static 'main/js/jquery-migrate-3.0.1.min.js' %}"></script>
<script src="{% static 'main/js/jquery-ui.js' %}"></script>
<script src="{% static 'main/js/popper.min.js' %}"></script> <script
src="{% static 'main/js/bootstrap.min.js' %}"></script> <script src="{%
static 'main/js/owl.carousel.min.js' %}"></script> <script src="{% static
'main/js/jquery.stellar.min.js' %}"></script> <script src="{% static
'main/js/jquery.countdown.min.js' %}"></script> <script src="{% static
'main/js/bootstrap-datepicker.min.js' %}"></script> <script src="{%
static 'main/js/jquery.easing.1.3.js' %}"></script> <script src="{% static
'main/js/aos.js' %}"></script>
<script src="{% static 'main/js/jquery.fancybox.min.js' %}"></script>
<script src="{% static 'main/js/jquery.sticky.js' %}"></script> <script
src="{% static 'main/js/main.js' %}"></script>
<script src="{% static 'admin/vendor/jquery/jquery.min.js' %}"></script>
<script src="{% static 'admin/vendor/bootstrap/js/bootstrap.bundle.min.js' %}"></script>
<script src="{% static 'admin/vendor/jquery-easing/jquery.easing.min.js' %}"></script>
<script src="{% static 'admin/js/ruang-admin.min.js' %}"></script>
</body>

</html>

```

CHAPTER 10

RESULTS/DISCUSSION

10.1 SYSTEM TEST

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing :

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

- Invalid Input : identified classes of invalid input must be rejected.
- Functions : identified functions must be exercised.
- Output : identified classes of application outputs must be exercised.
- Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

10.1.1 TEST CASES

Test case1 for Login form:

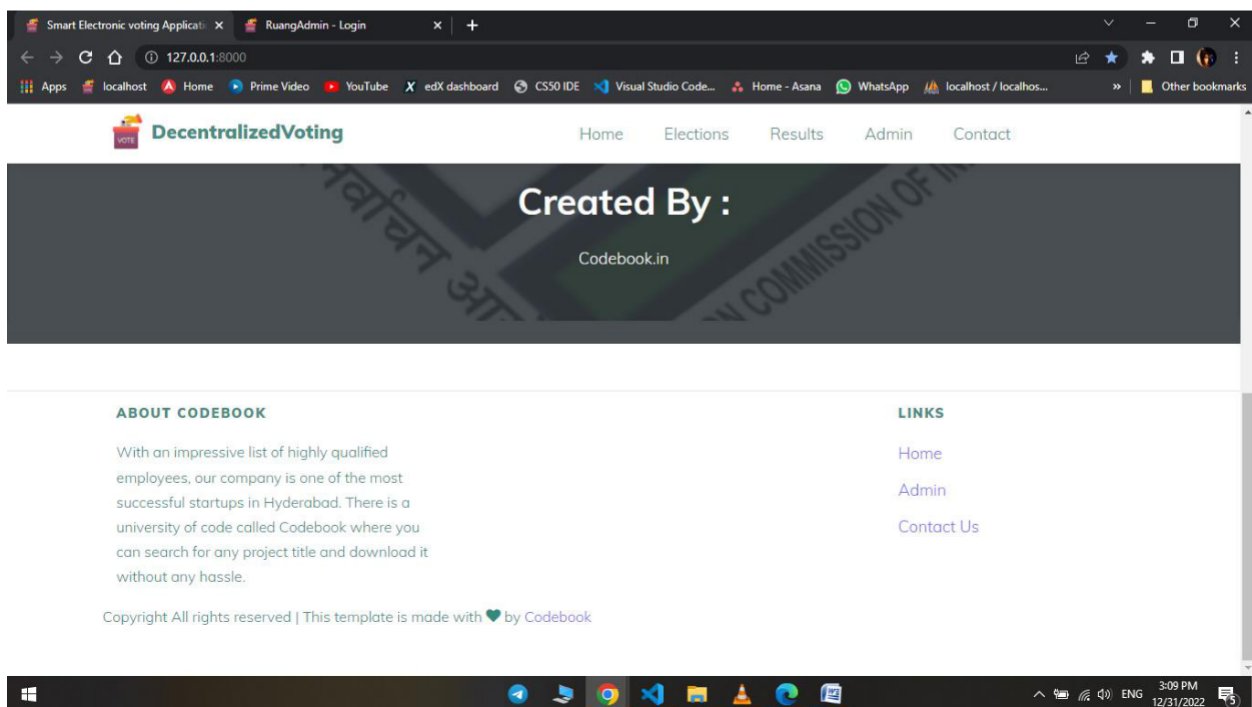
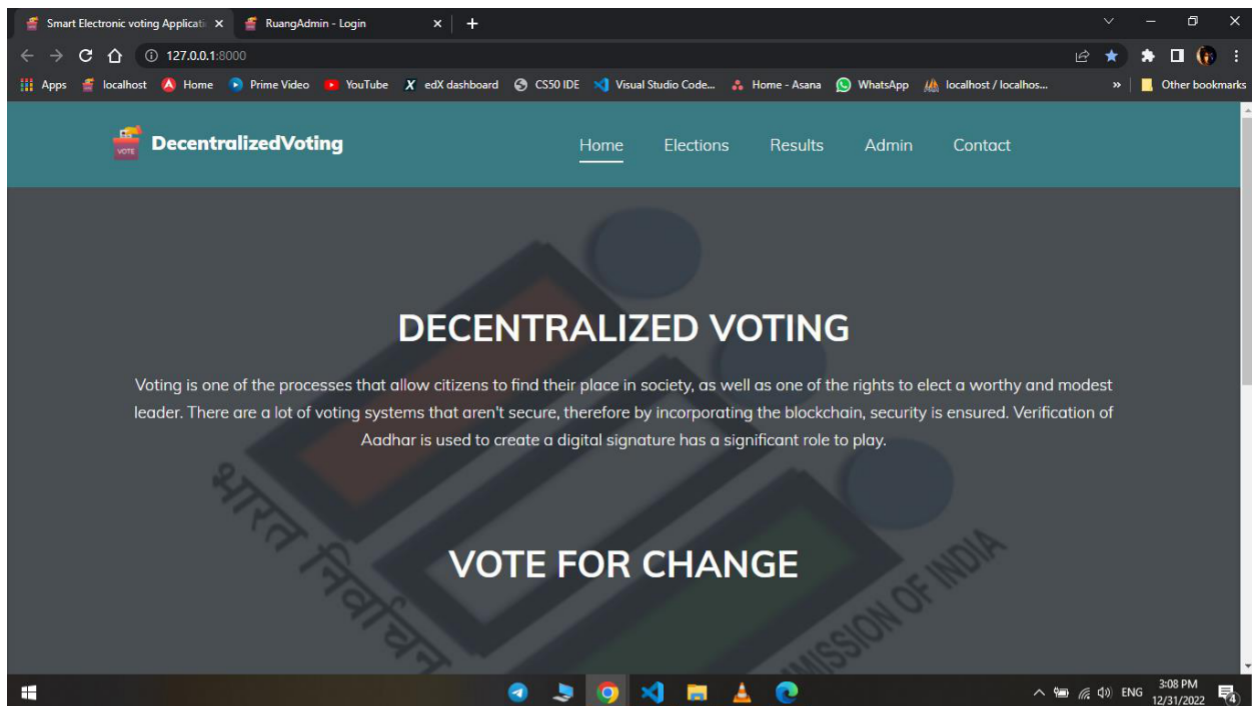
FUNCTION:	LOGIN
EXPECTED RESULTS:	Should Validate the user and check his existence in database
ACTUAL RESULTS:	Validate the user and checking the user against the database
LOW PRIORITY	No
HIGH PRIORITY	Yes

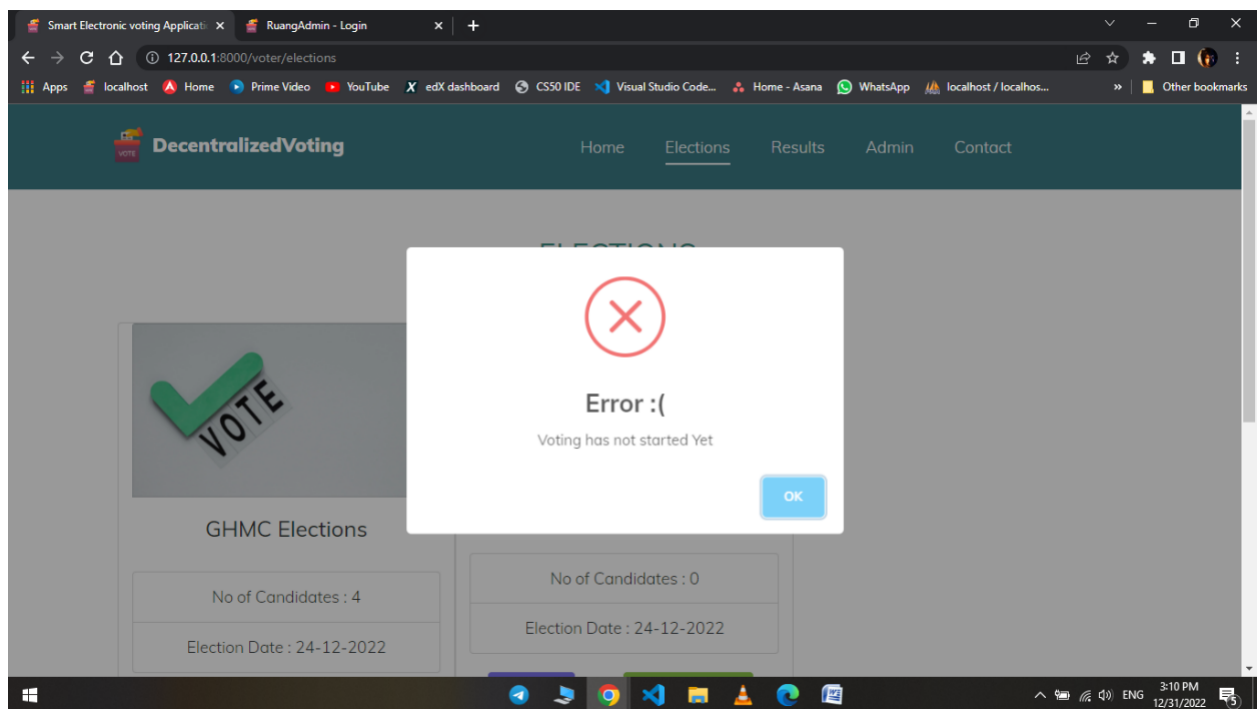
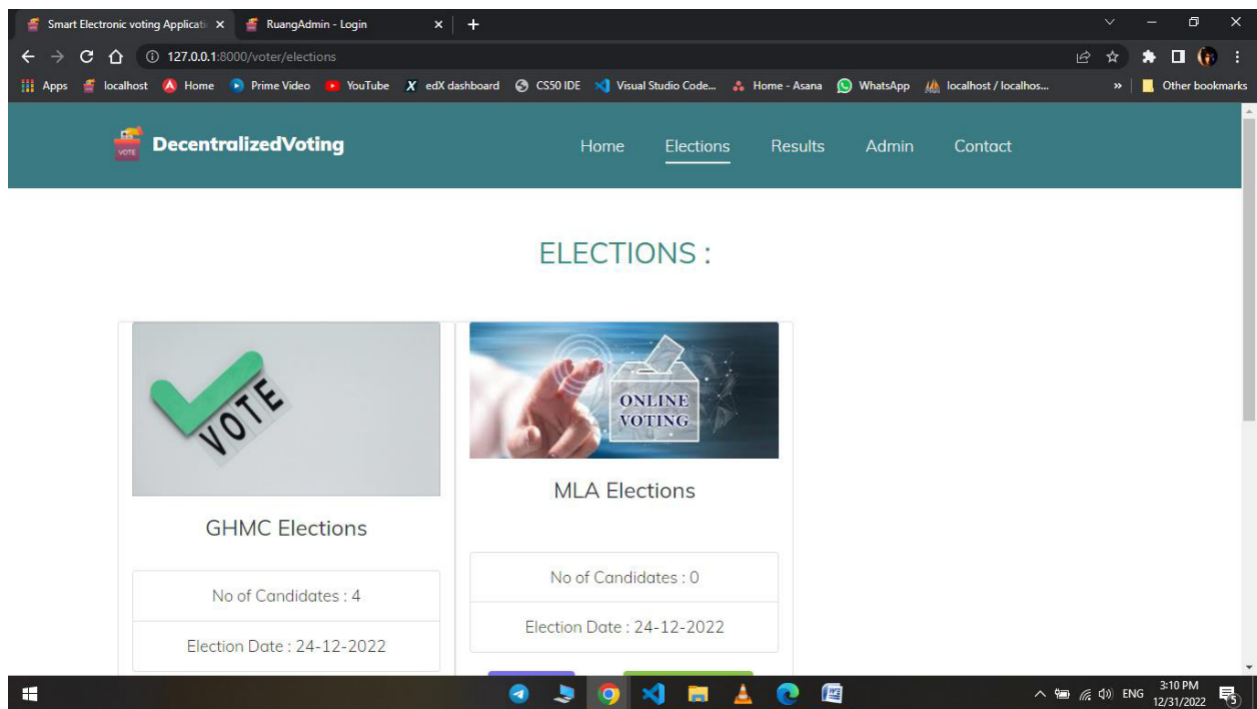
Test case2:

Test case for User Registration form:

FUNCTION	USER REGISTRATION
EXPECTED RESULTS:	Should check if all the fields are filled by the user and saving the user to database.
ACTUAL RESULTS:	Checking whether all the fields are filled by user or not through validations and saving user.
LOW PRIORITY	No
HIGH PRIORITY	Yes

10.2 SCREENSHOTS





Smart Electronic voting Application | RuangAdmin - Login

127.0.0.1:8000/voter/voting/1

DecentralizedVoting

Home Elections Results Admin Contact

Election Name : GHMC Elections

Constituency : General Election

Information About The candidates :

Candidate Name	Party Name	Symbol	Action
Rama Krishan Pawar	Janasena		VOTE
Sri Owaisi Asaduddin	TRS		VOTE
Dr. Gaddam Ranjith Reddy	TDP		VOTE

Smart Electronic voting Application | RuangAdmin - Login

127.0.0.1:8000/voter/register/5

DecentralizedVoting

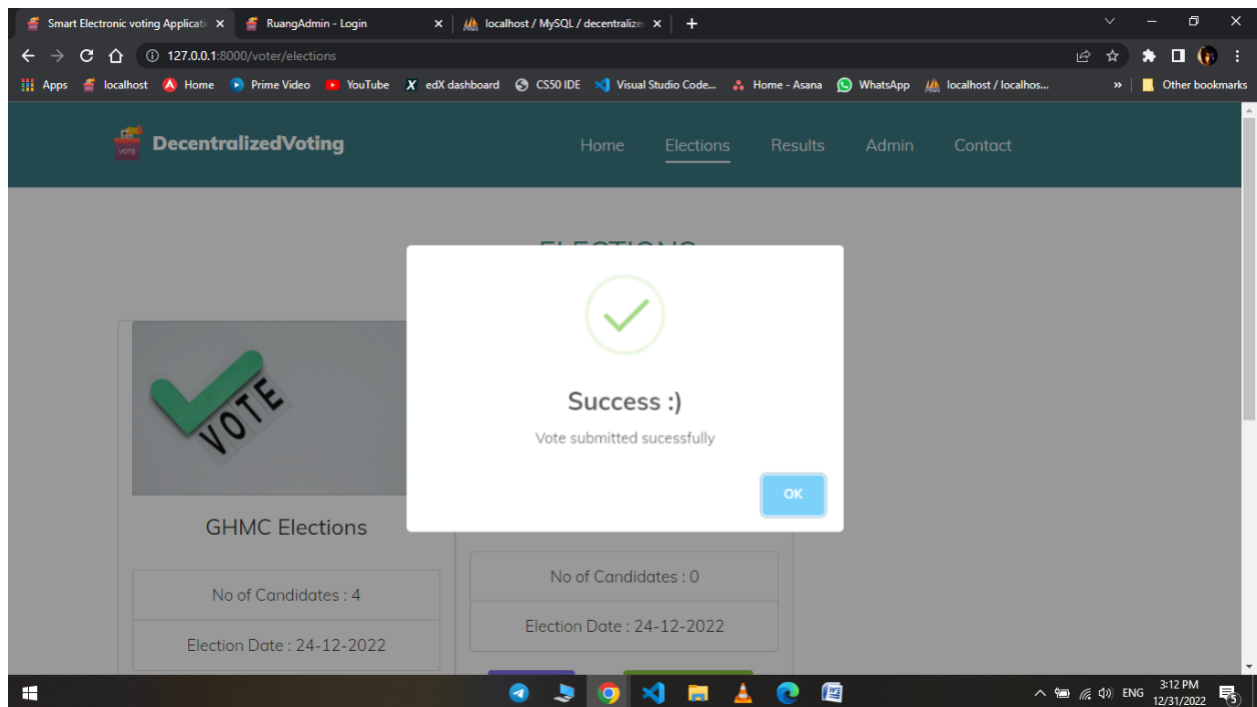
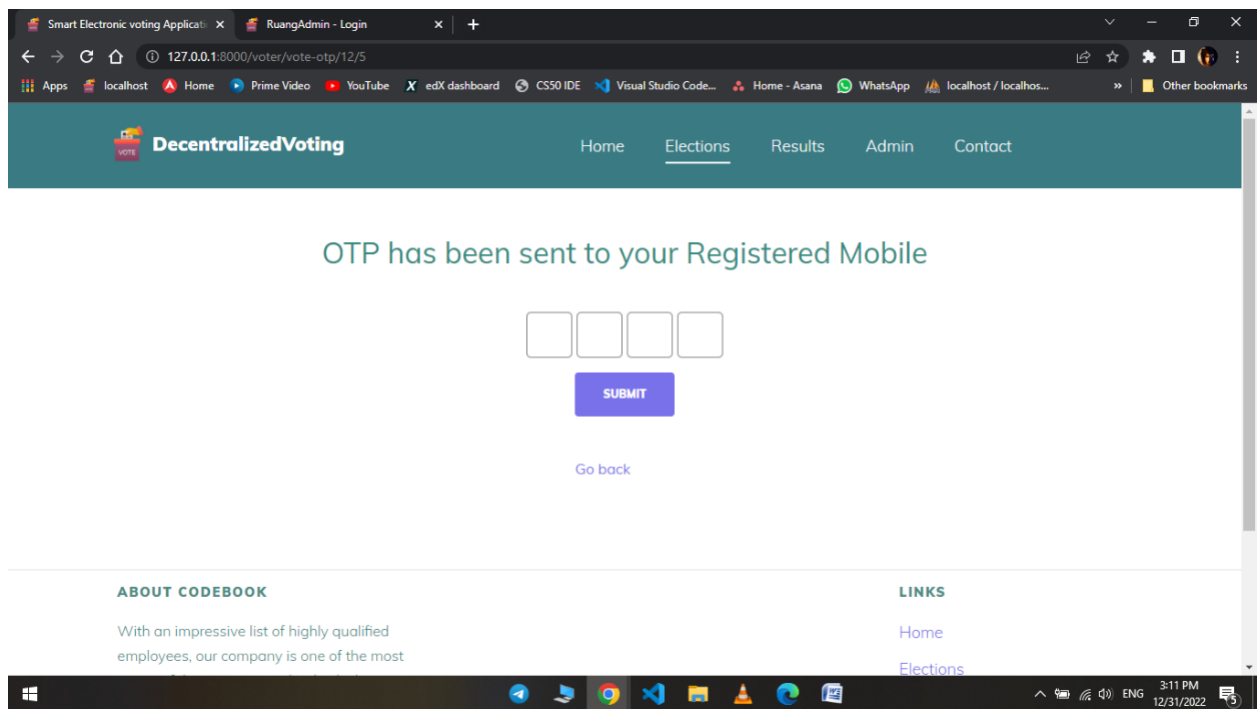
Home Elections Results Admin Contact

Voter Verification form

Phone number :

Aadhar Number:

[SUBMIT](#)



Smart Electronic voting Application | RuangAdmin - Login | localhost / MySQL / decentralize

127.0.0.1:8000/voter/voter-results

DecentralizedVoting

Home Elections Results Admin Contact

The Below Table Showing The Current Situation In All Elections :

Sno.	Election Name	Constituency	Date	Results	Action
2	Mayor Elections 2022	 General Election	Dec. 30, 2022	Declared	VIEW RESULT

<- Previous Page Next Page ->

ABOUT CODEBOOK

With an impressive list of highly qualified employees, our company is one of the most

LINKS

[Home](#)

[Elections](#)

WhatsApp | Smart Electronic voting Application

127.0.0.1:8000/voter/voter-view-result/2

DecentralizedVoting

Home Elections Results Admin Contact

Winner of the Election is "Kiran" from Congress party.

Results Verified Using Blockchain

Total Votes: 9

Pie Chart Analysis



Legend: Validated (Green), Invalid (Red)

Voter Aadhar	Block 1	Block 2	Block 3	Status
--------------	---------	---------	---------	--------

WhatsApp Smart Electronic voting Application 127.0.0.1:8000/voter/voter-view-result/2

DecentralizedVoting Home Elections Results Admin Contact



Voter Aadhar	Block 1	Block 2	Block 3	Status
8458745874588547	050c5c21729b793a7214 4096546943fe666d2e89	0dfcb3a188372ce2e88ce 42725cc2bcfe0d5399b70	35a39748290096de21fe 3a6fda78cafad8dd4aa57	Valid
8745214587458745	6c35ac9de912851d9739 ce7d396b02827db910d5	cd3f228f9d5ccefd3a1d82 1703f5ea51a6c134de1d	f29c902529909cb3fb865 8263750f5858d6b4db67	Valid
8663251478521458	dd45f84d1086bad5733b 24c4143df6448ee598de1	887171ee0697157de1ea efa0ab85770dde8d6409f	f2ab98694f951caed1c2a cab3b26e6b1a24d723a0	Invalid
8512665478512544	b2f285beda36d96538ed bda9be761560c39d7fb6	8eba5495d438afb15044 15f8a171d07059d0f3d0d	917b52a9ded6b05dbcb6 bde8f5af5dbb48d32a49	Valid

12:58 PM 1/4/2023

Smart Electronic voting Application RuangAdmin - Login localhost / MySQL / decentralize 127.0.0.1:8000/contact

DecentralizedVoting Home Elections Results Admin Contact

Message Us

Any Queries Regarding To This Website , Please Free To Message Us.

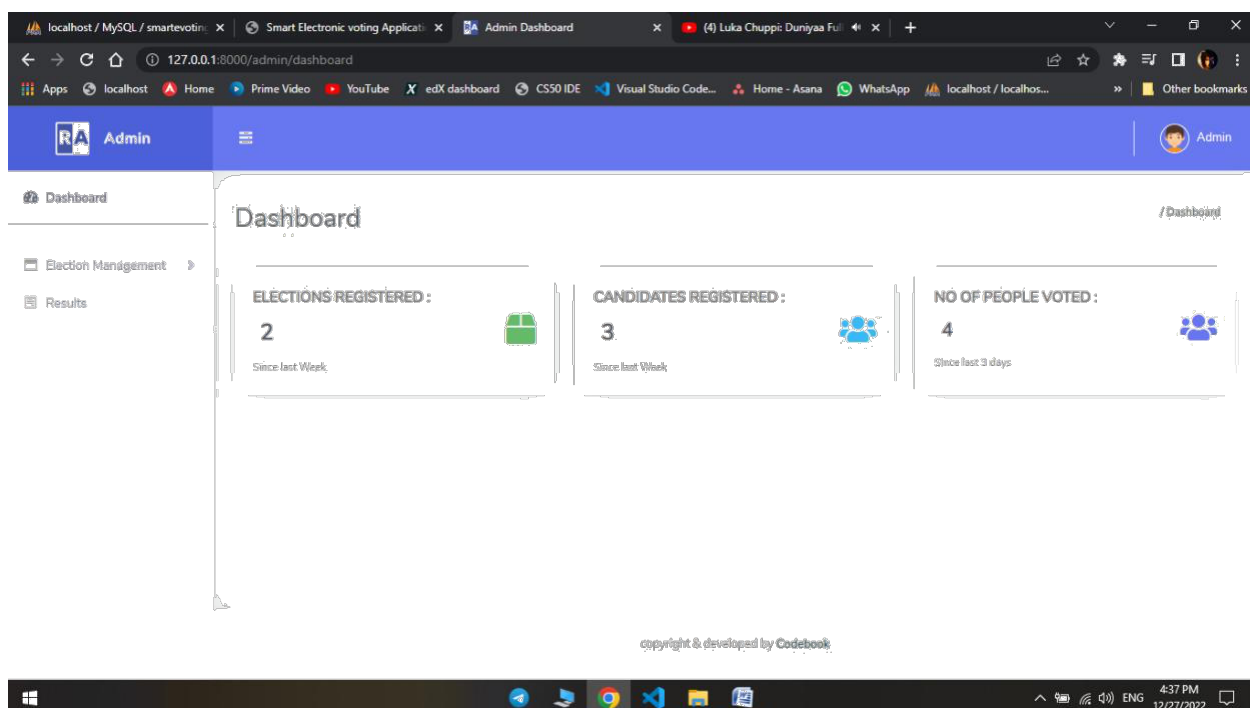
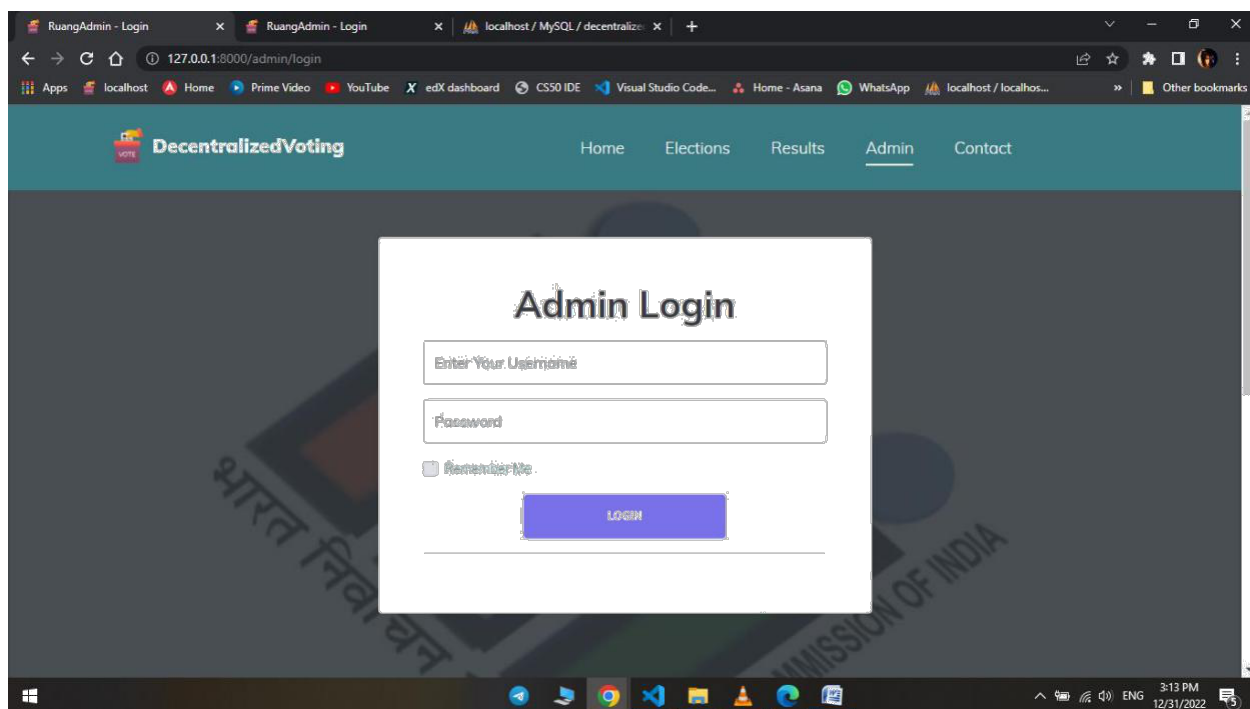
First name Last name

Subject

Email

Write your message here.

3:13 PM 12/31/2022



localhost / MySQL / smartevoting : x Smart Electronic voting Appli... x Admin-addelection x (4) Luka Chuppi Duniyaa Fu... x +

127.0.0.1:8000/admin/add-election

Apps localhost Home Prime Video YouTube edX dashboard CS50 IDE Visual Studio Code... Home - Asana WhatsApp localhost / localhos... Other bookmarks

Results

Election Registration Form :

Election Name : Head of Election :

Election-Picture : No file chosen Election-Date :

Constituency : Area to Conduct Election:

Address (Exact):

City : State : Zip :

4:38 PM 12/27/2022

localhost / MySQL / smartevoting : x Smart Electronic voting Appli... x Admin - addcandidate x (4) Luka Chuppi Duniyaa Fu... x +

127.0.0.1:8000/admin/add-candidate

Apps localhost Home Prime Video YouTube edX dashboard CS50 IDE Visual Studio Code... Home - Asana WhatsApp localhost / localhos... Other bookmarks

Admin

Dashboard

Election Management >

- Add Election
- Add Candidate**
- View Elections

Results

Add Candidates To Elections

Election Management / Add Candidates To Elections

Add your candidate to the elections by filling out the following form:

Candidate Full Name : Party Name :

Election Name (participating in) : Party Symbol : No file chosen

4:38 PM 12/27/2022

localhost / MySQL / smartevoting | Smart Electronic voting Application | Admin - view-elections | (4) Luka Chuppi Duniyaa Full |

127.0.0.1:8000/admin/view-elections

Apps | localhost | Home | Prime Video | YouTube | edX dashboard | CS50 IDE | Visual Studio Code... | Home - Asana | WhatsApp | localhost / localhos... | Other bookmarks

RA Admin

Dashboard

Election Management >

- Add Election
- Add Candidate
- View Elections**

Results

Elections

Election Management / Elections

Sno.	Election Name	Picture	Head of Election	Election Date	Constituency	Action
1	GHMC Elections		GHMC Elections	Dec. 24, 2022	General Election	View
2	Mayor Elections 2022		Surya Kumar Yadav	Dec. 27, 2022	General Election	View

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localhost / MySQL / smartevoting | Smart Electronic voting Application | Admin - view-elections | (4) Luka Chuppi Duniyaa Full |

127.0.0.1:8000/admin/view-candidates/1

Apps | localhost | Home | Prime Video | YouTube | edX dashboard | CS50 IDE | Visual Studio Code... | Home - Asana | WhatsApp | localhost / localhos... | Other bookmarks

RA Admin

Dashboard

Election Management >

- Add Election
- Add Candidate
- View Elections**

Results

Candidates for Election: GHMC Elections

Election Management / Elections

Sno.	Candidate Name	Party Name	Symbol
1	Rajeev Reddy	Congress	
2	Krishna Kanth	TRS	

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localhost / MySQL / smartevoting: x Smart Electronic voting Applica: x Admin - results x (4) Luka Chuppi: Duniyaa Fu: x +

127.0.0.1:8000/admin/results

Apps localhost Home Prime Video YouTube edX dashboard CS50 IDE Visual Studio Code... Home - Asana WhatsApp localhost / localhos... Other bookmarks

RA Admin

Dashboard

Election Management >

Results

Results Table

/ Results

Sno.	Election Name	Picture	Head of Election	Election Date	Constituency	Action
1	GHMC Elections		GHMC Elections	Dec. 24, 2022	General Election	Verify Result
2	Mayor Elections 2022		Surya Kumar Yadav	Dec. 27, 2022	General Election	Verify Result

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localhost / MySQL / smartevoting: x Smart Electronic voting Applica: x Admin - results x (4) Sooraj Dooba Hain: FULL: x +

127.0.0.1:8000/admin/verify-results/1

Apps localhost Home Prime Video YouTube edX dashboard CS50 IDE Visual Studio Code... Home - Asana WhatsApp localhost / localhos... Other bookmarks

RA Admin

Dashboard

Election Management >

Results

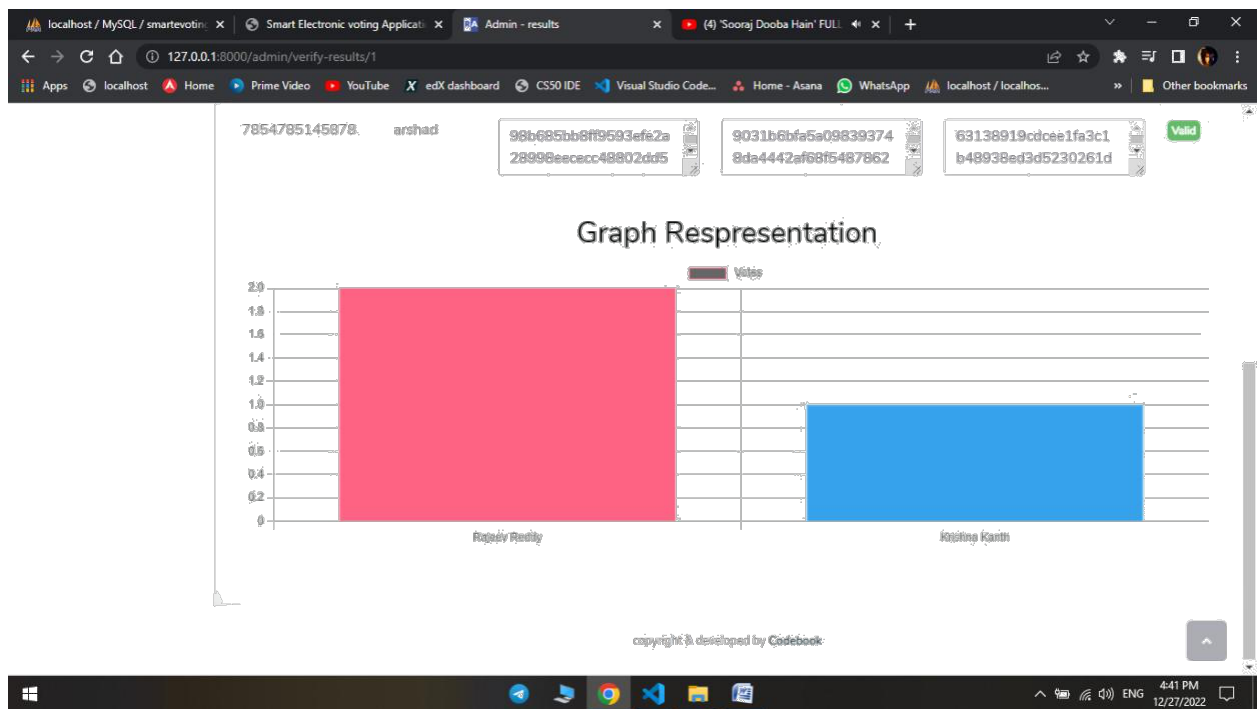
Results Verified Using Blockchain

Validated: 2 / 3 votes

Invalid: 1

Total: 3

Voter Aadhar	Voter Name	Block 1	Block 2	Block 3	Status
62378514965478	Rajesh	590c910e02cb45e79d030fdc15a4a34d410d36	33b6a27f0e55c66ab01816d6d7b1855078beb	534cf3dc57afc6639ec072e0be93bc53428cb4a	Valid
4587412365985	kiran kumar	aa69cf1a8f77f13b8d0fb97377a0e27f1328a55	93042be3ff6f9c86acd652c35290bf39123a718	304c25020d952111320d3611da4b00958923	Invalid
7854785145878	arshad	98b685bb8ff9593efe2a28998eececc48802dd5	9031b6bfa5a098393748da442af68f5487862	63138919cdcee1fa3c1b48938ed3d5230261d	Valid



CHAPTER 11

CONCLUSION

11.1 CONCLUSION

The proposed solution is to create an online voting system based on the Ethereum Blockchain and based on voter identification given by UIDAI with face and fingerprint recognition, which attempts to address all of the shortcomings of the old voting system. The existing system needs the actual presence of the voter which is not convenient and chances of vote tampering is more. Also, the transparency of the entire system is not maintained. Our system uses blockchain technology which helps in maintaining the transparency, accuracy, verifiability, ease, security, of the system. The suggested method uses double verification for increased security. Because the system is an online application, users may vote from anywhere. The voter does not need to go to a polling location to vote, allowing a larger number of residents to participate. The Ethereum Blockchain will be used to safeguard the data in the proposed system.

CHAPTER 12

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